



Flow-Based Market Coupling

Dr. ir. Pieter Schavemaker
E-Bridge Consulting B.V.

NTNU, Trondheim, Norway

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Questions?

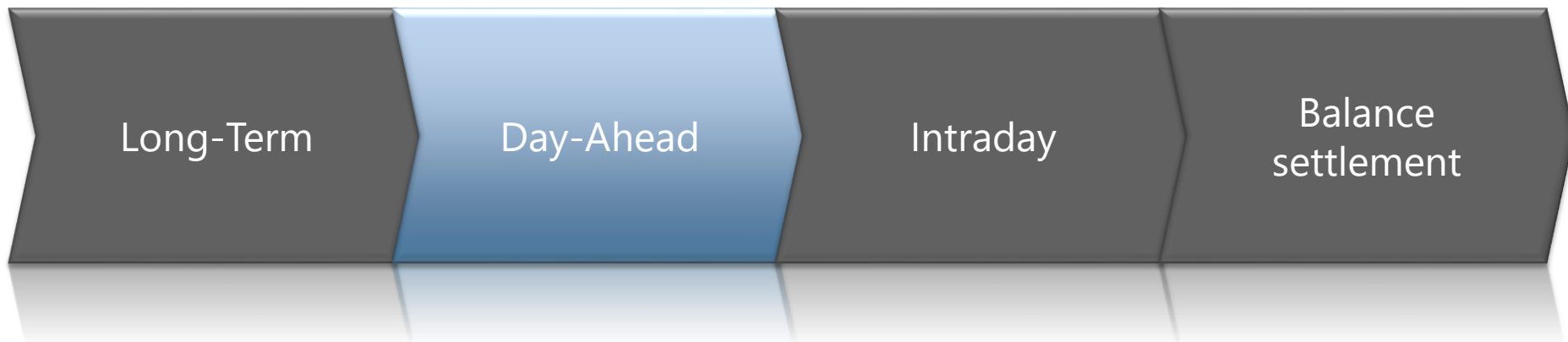
- Always welcome, do not hesitate to ask!



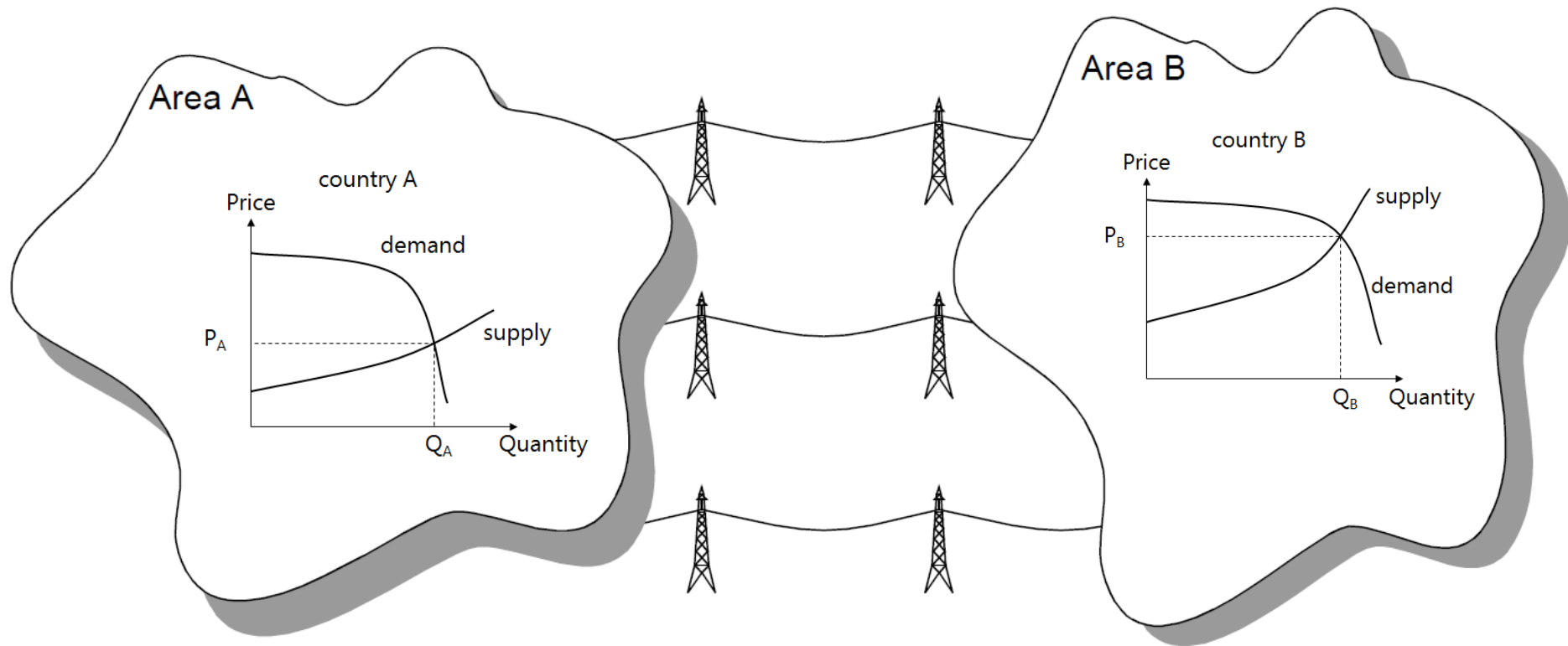


Market Coupling

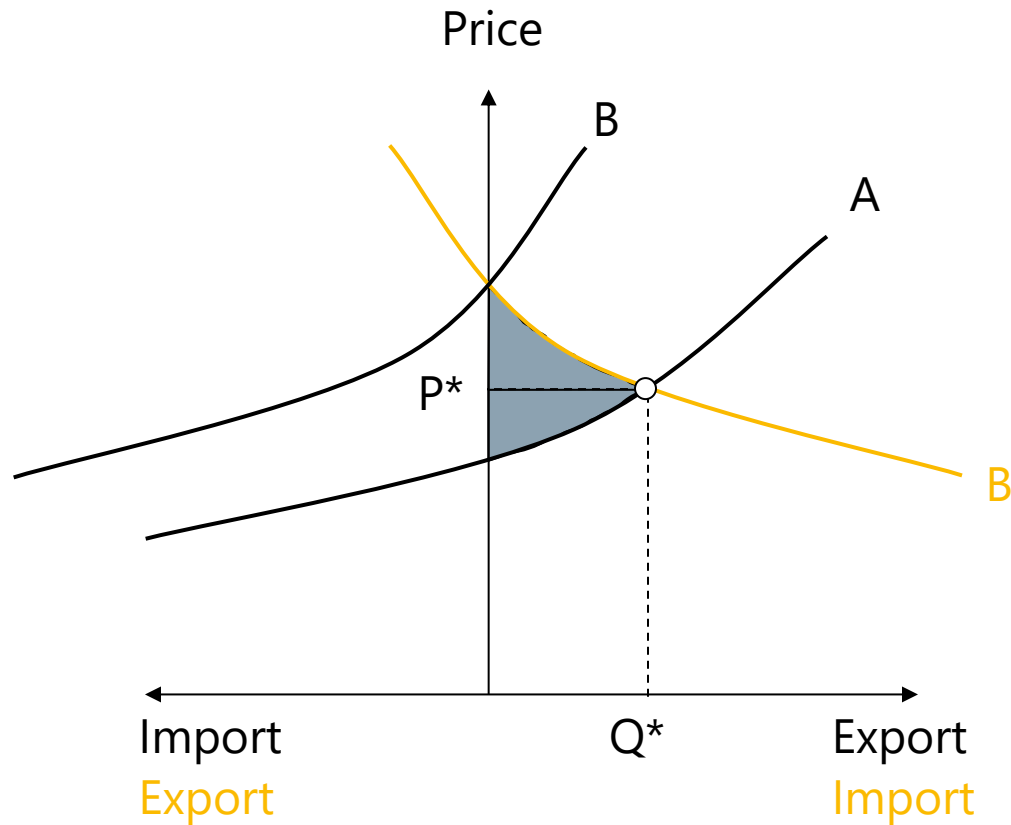
Market segments



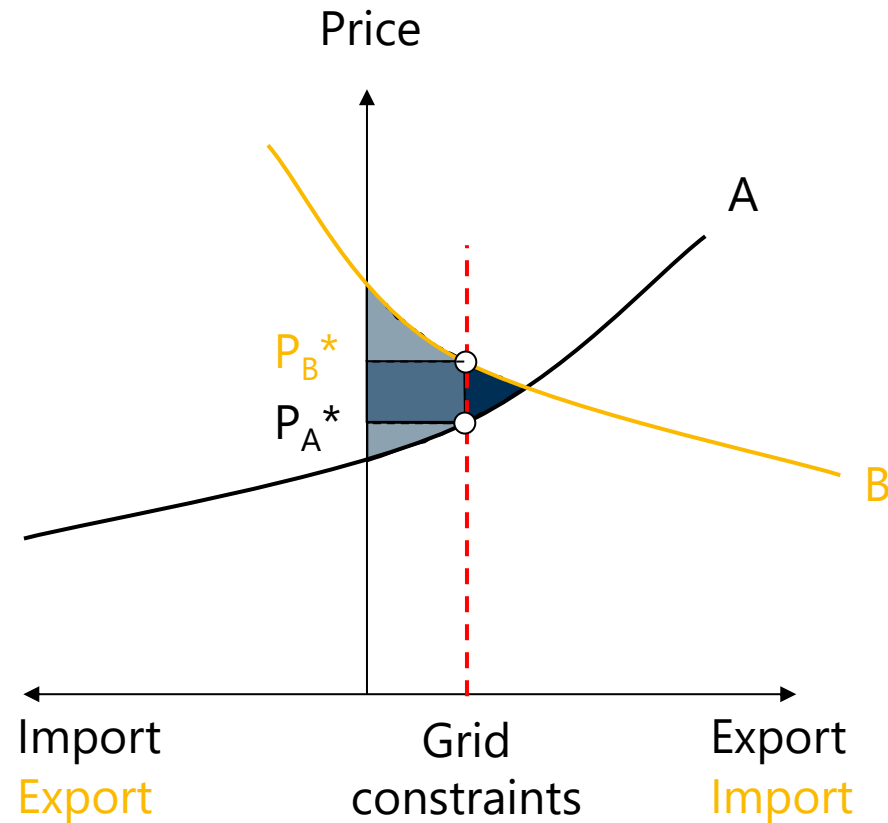
Market coupling



Increase of Social Welfare by Import and Export



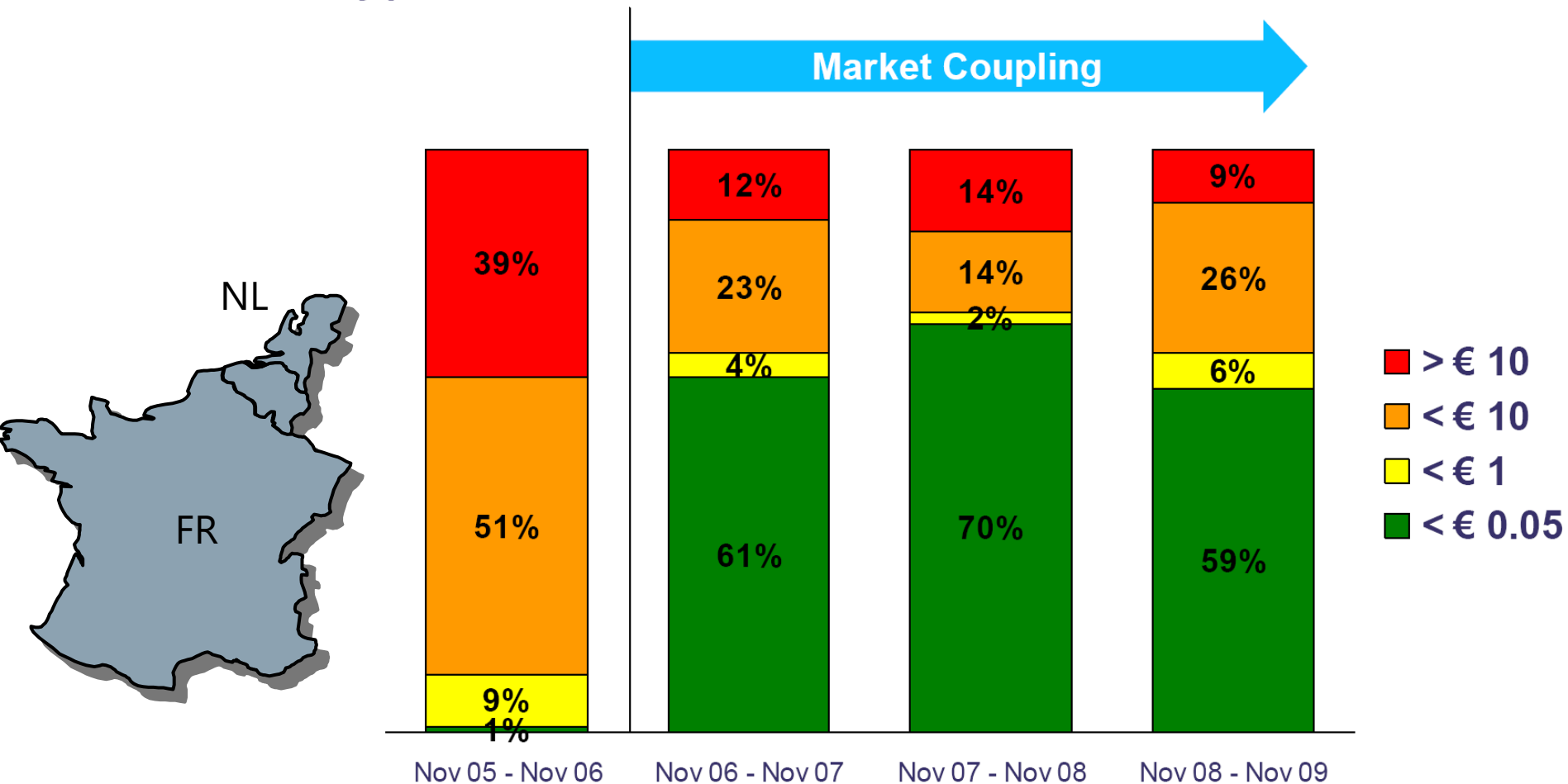
But how about the grid?



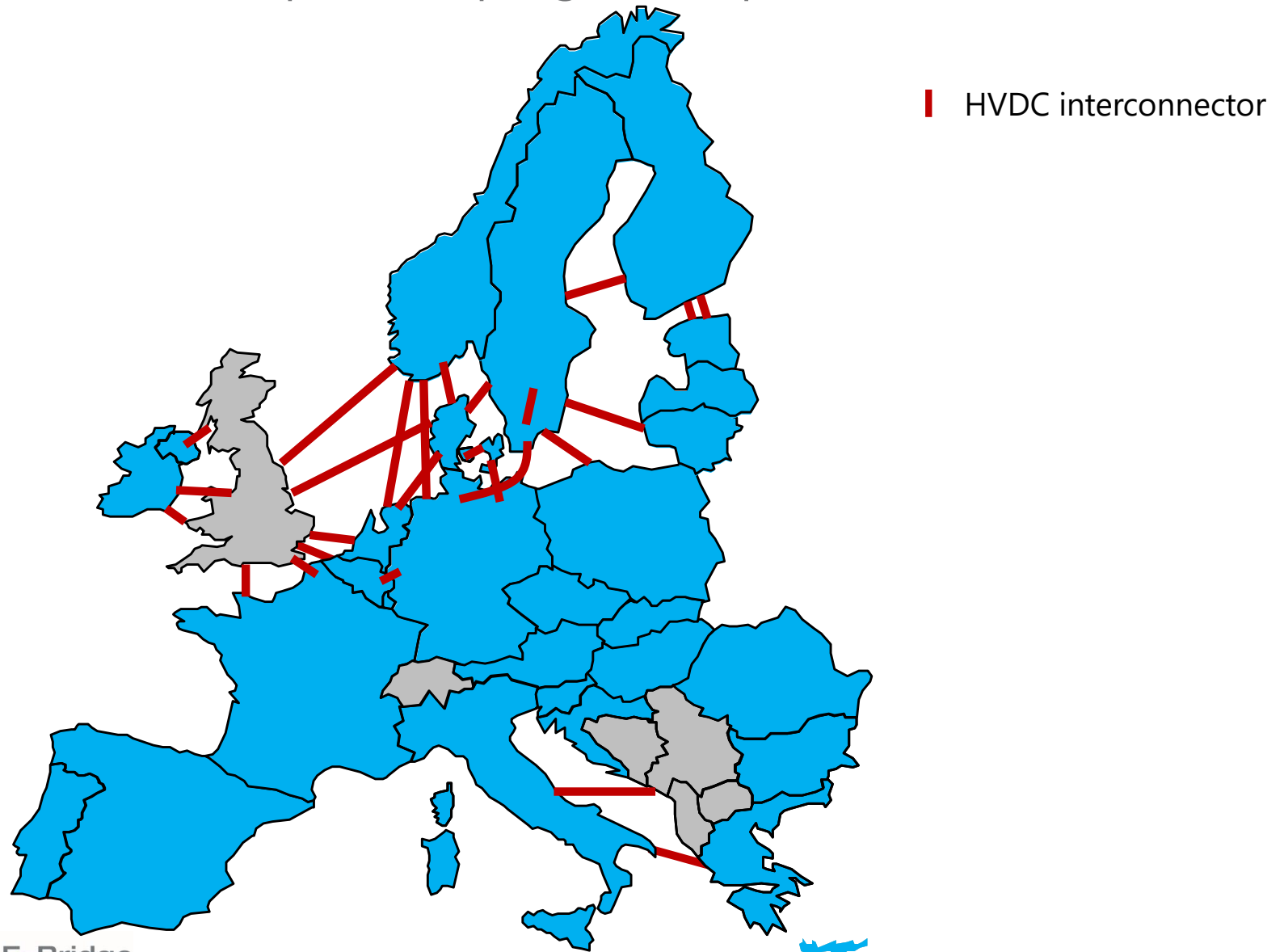
Social welfare = consumer surplus + producer surplus +
congestion revenue

Price difference between NL-FR With a Market coupling of FR-BE-NL as of Nov 2006

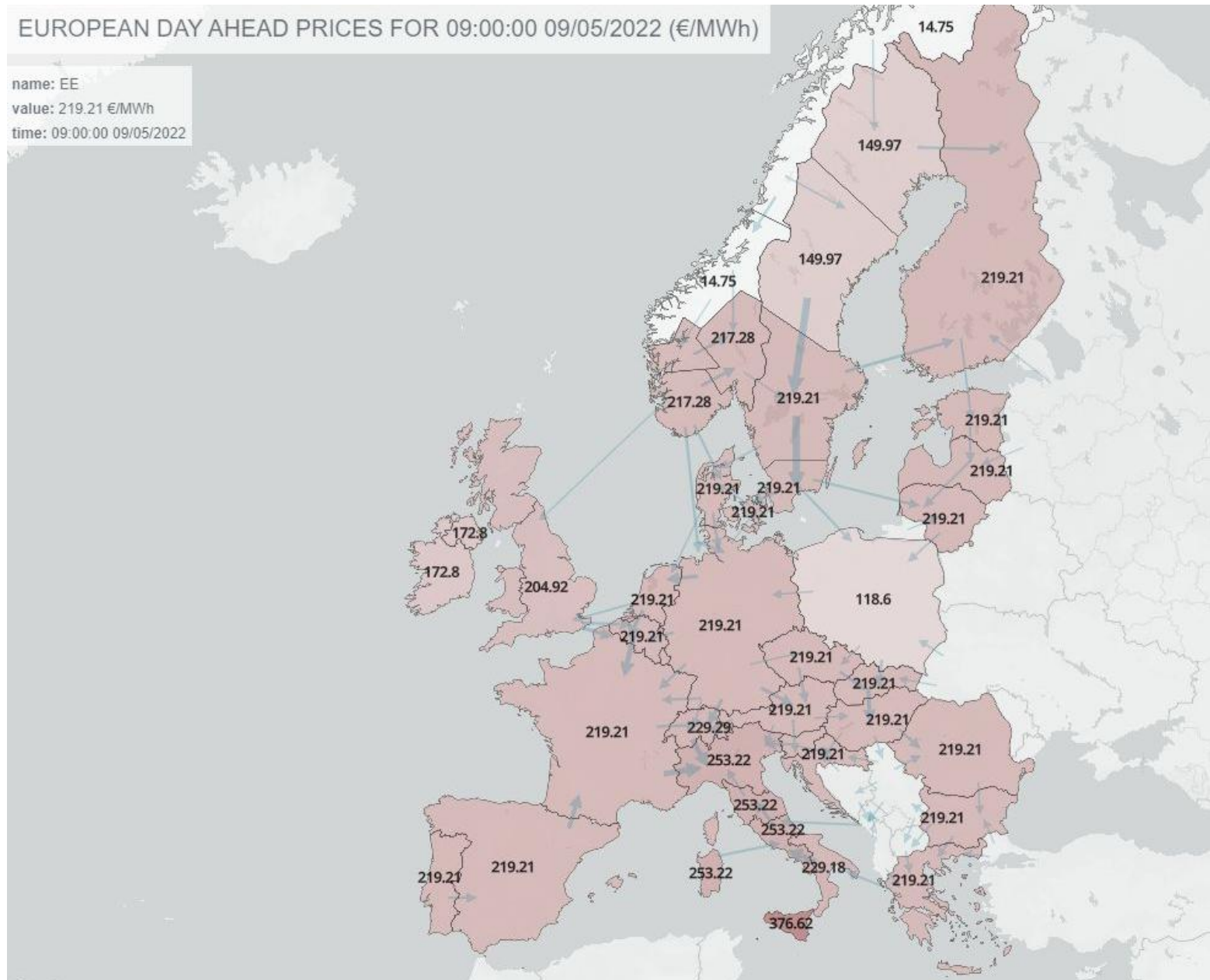
Hourly price difference, €/MWh



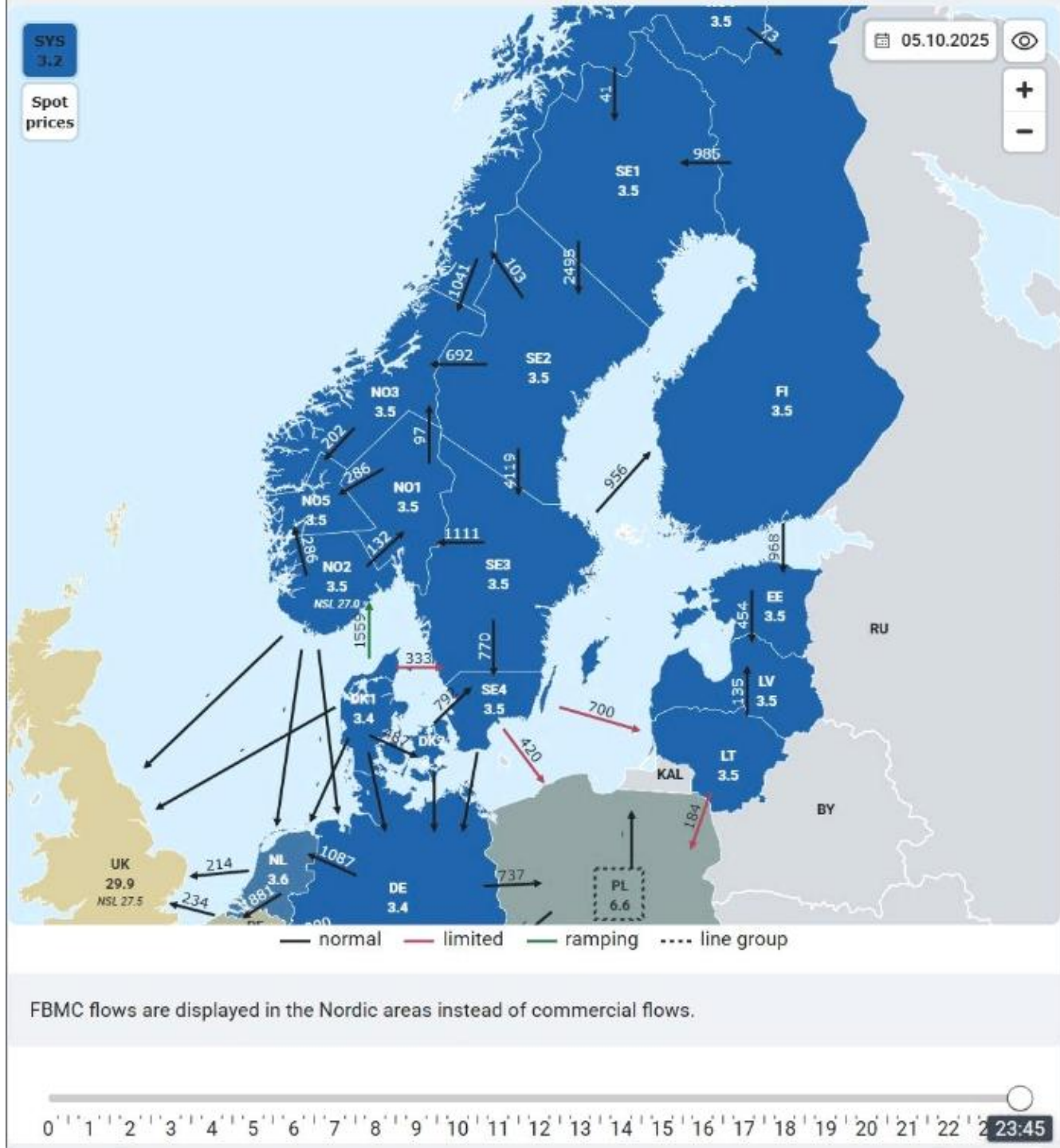
Current status of price coupling in Europe



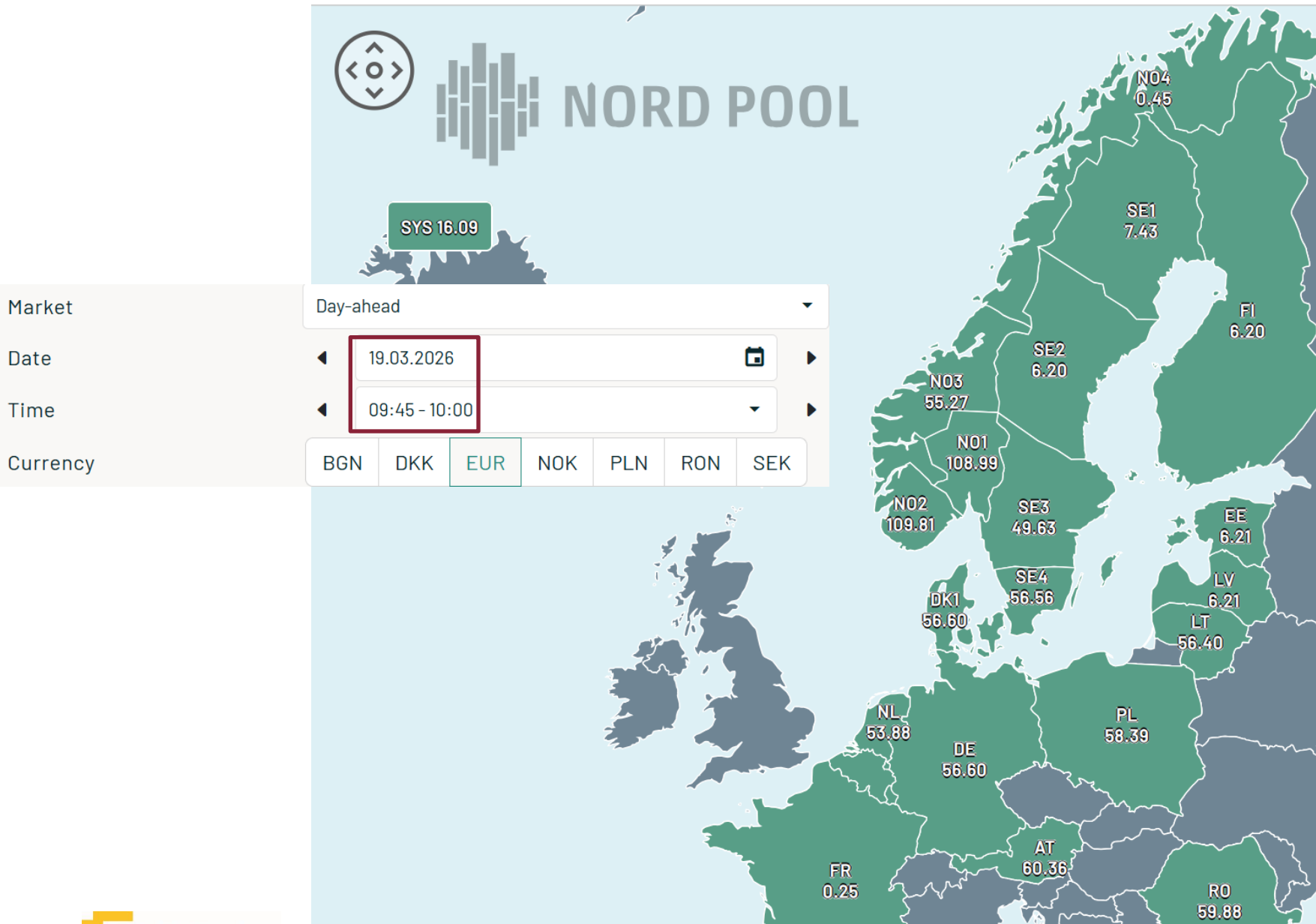
Market Coupling on May 9, 2022



Market Coupling on October 5, 2025

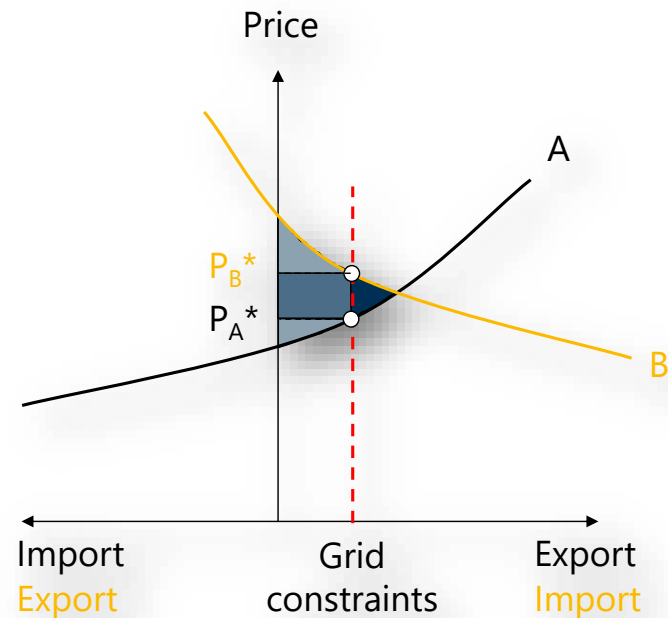
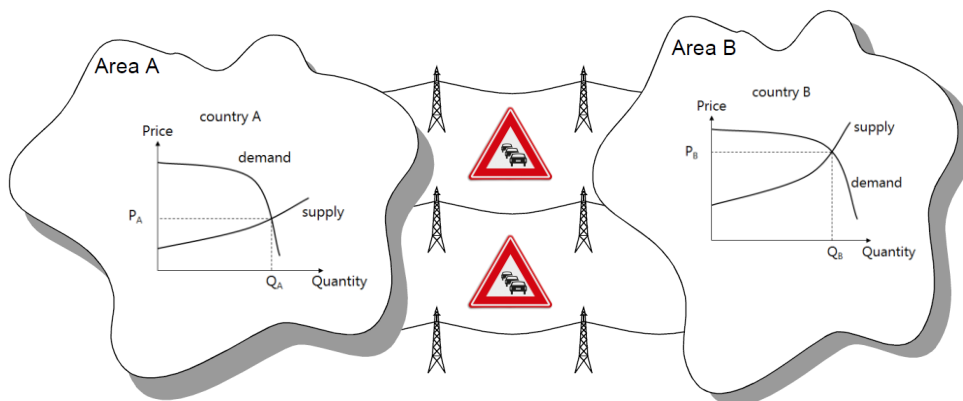


Example prices Nord Pool



Market Coupling: a constrained optimization problem

- All the bids of the bidding areas are brought together in order to be matched by a centralized algorithm
- Objective function: Maximize social welfare
- Control variables: Net positions
- Subject to: $\sum \text{net positions} = 0$
Grid constraints





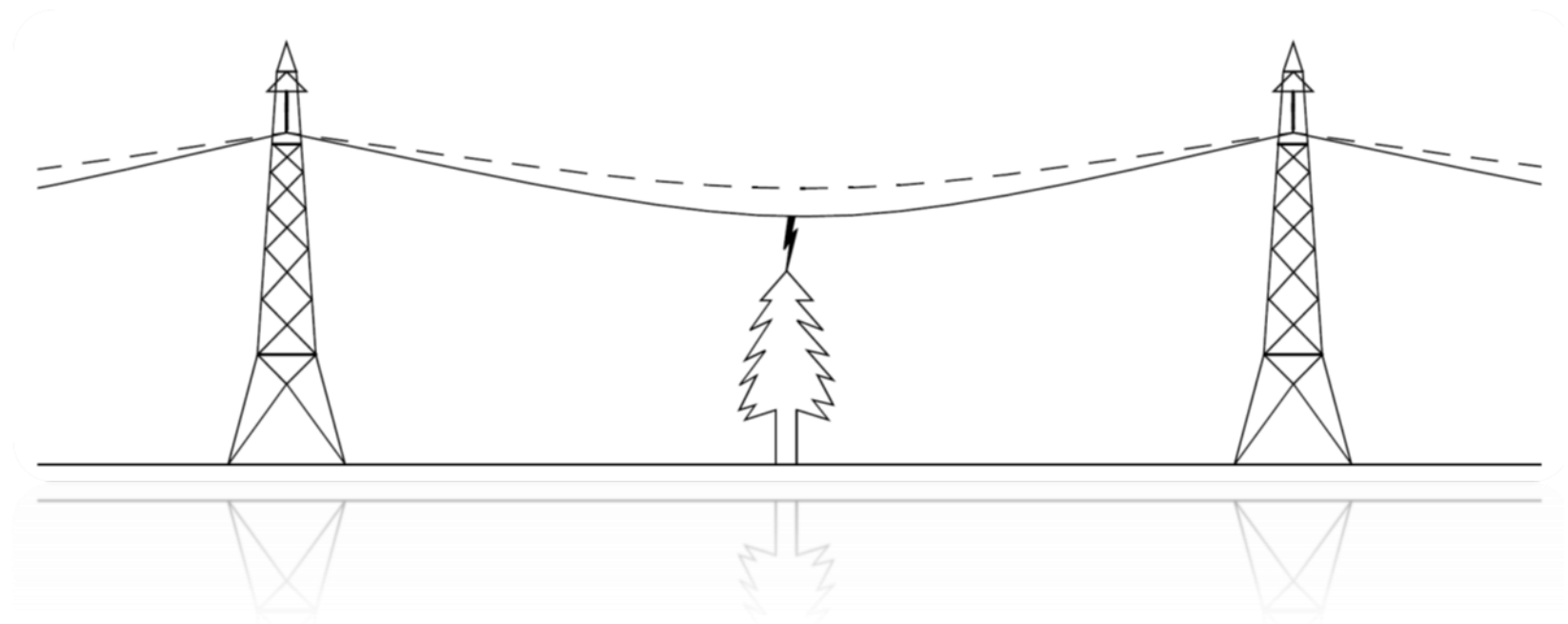
Congestion management

What is congestion?

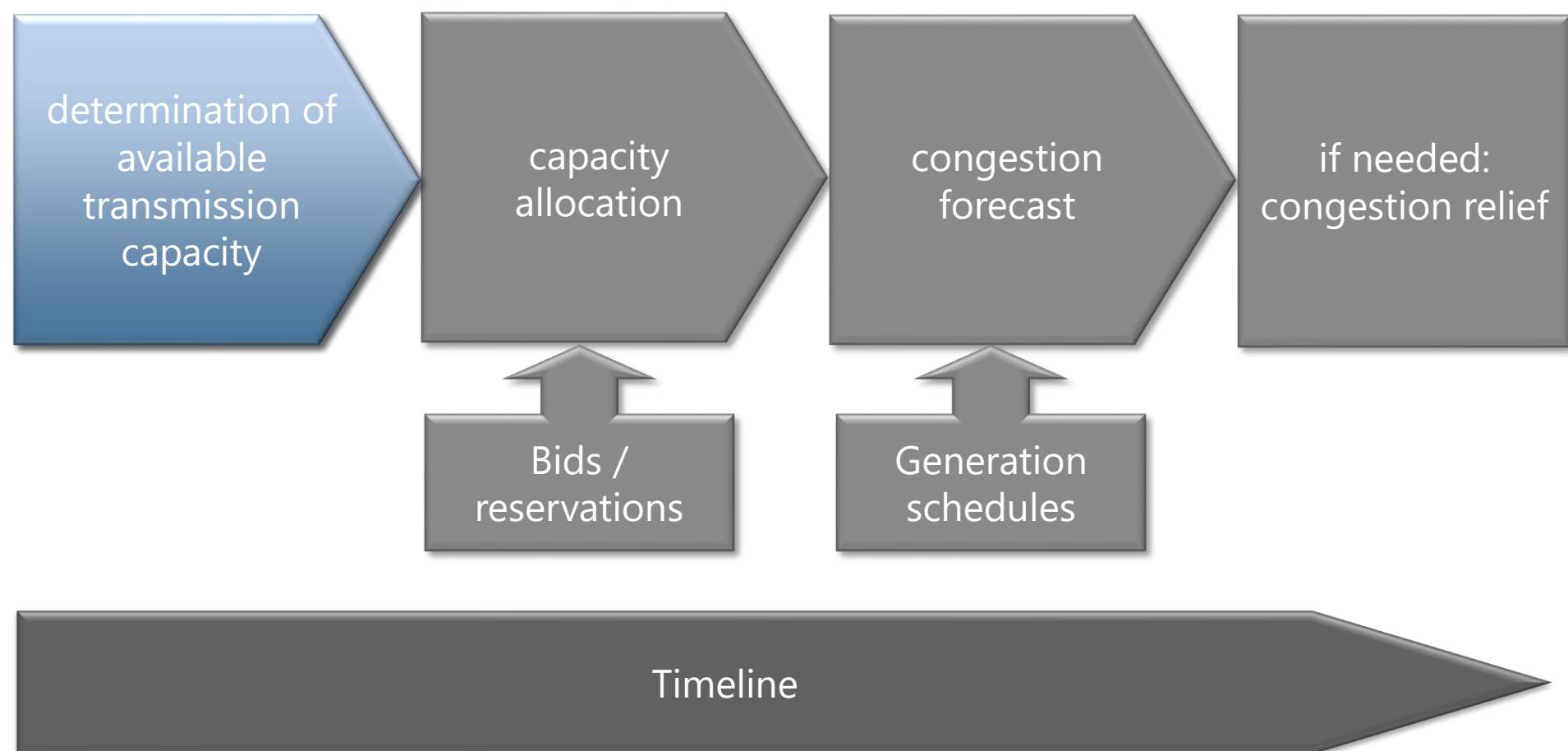


What is congestion?

- commercial: more capacity requested by the market than is available
- physical: overloaded transmission lines leading to outages



Congestion management in the broadest sense



Guideline on Capacity Allocation and Congestion Management (CACM GL)



L 197/24

EN

Official Journal of the European Union

25.7.2015

COMMISSION REGULATION (EU) 2015/1222

of 24 July 2015

establishing a guideline on capacity allocation and congestion management

(Text with EEA relevance)

- Article 20.2:
'No later than 10 months after the approval of the proposal for a capacity calculation region in accordance with Article 15(1), all TSOs in each capacity calculation region shall submit a proposal for a common coordinated capacity calculation methodology within the respective region.'
- The ACER decision on the TSO's proposal for the determination of Capacity Calculation Regions dates November 17, 2016

CACM GL: cross-zonal capacity (i.e. capacity between bidding zones)

L 197/24

EN

Official Journal of the European Union

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COMMISSION REGULATION (EU) 2015/1222
of 24 July 2015
establishing a guideline on capacity allocation and congestion management
(Text with EEA relevance)

'There are two permissible approaches when calculating cross-zonal capacity: flow-based or based on coordinated net transmission capacity. The flow-based approach should be used as a primary approach for day-ahead and intraday capacity calculation where cross-zonal capacity between bidding zones is highly interdependent.'

...

'The coordinated net transmission capacity approach should only be applied in regions where cross-zonal capacity is less interdependent and it can be shown that the flow-based approach would not bring added value.'

Capacity calculation regions and FB

1. Nordic

2. Hansa

3. Core

4. Italy North

5. Greece-Italy (GRIT)

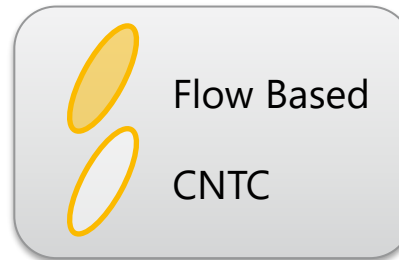
6. South-West Europe (SWE)

7. Ireland and United Kingdom (IU)

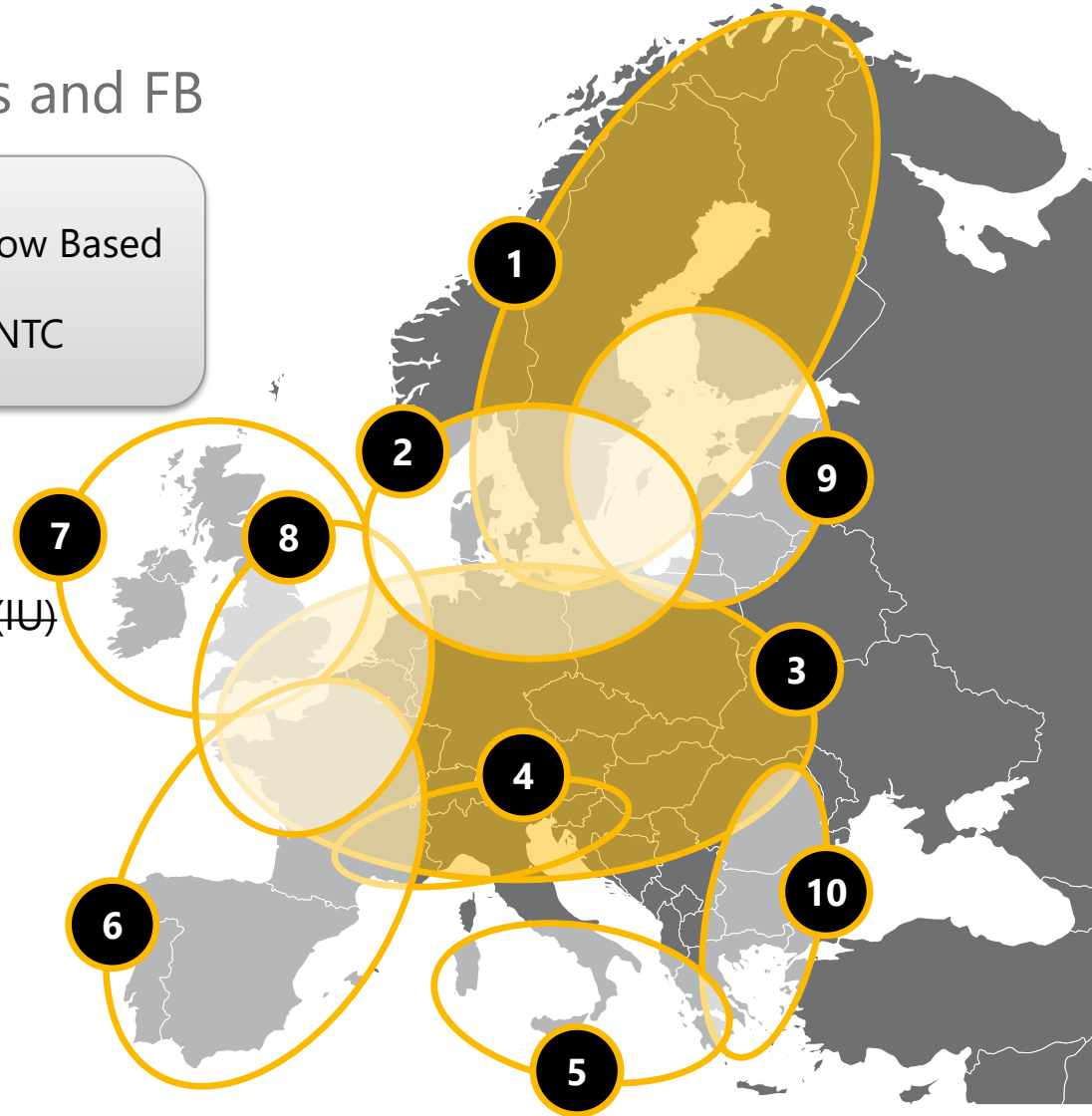
8. Channel

9. Baltic

10. South-East Europe (SEE)



Central Europe



Market Coupling: a constrained optimization problem

- All the bids of the bidding areas are brought together in order to be matched by a centralized algorithm

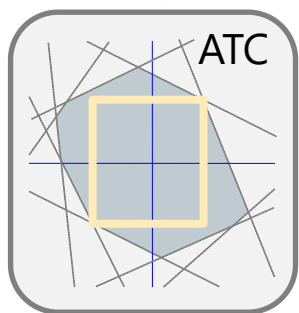
• Objective function: Maximize social welfare

• Control variables: Net positions

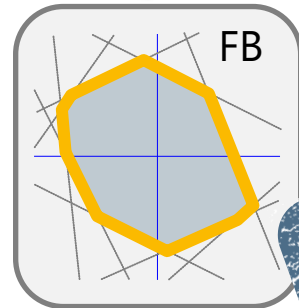
• Subject to: $\sum \text{net positions} = 0$

Grid constraints

Others



CWE



Determination
of available
transmission
capacity

capacity
allocation

Bids /
reservations



As of May 21, 2015!



Flow Based in Europe

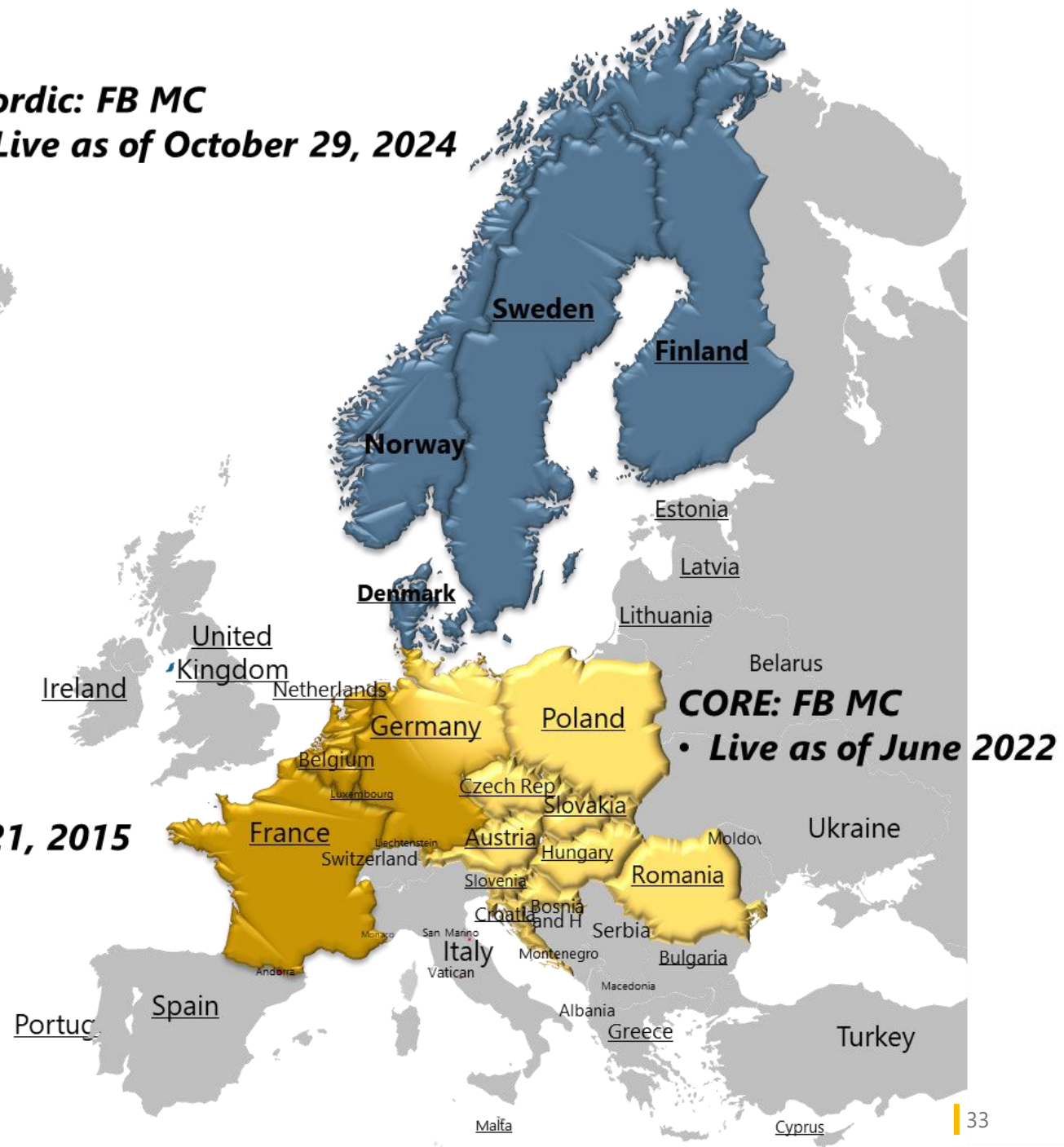
FB in Europe

Nordic: FB MC

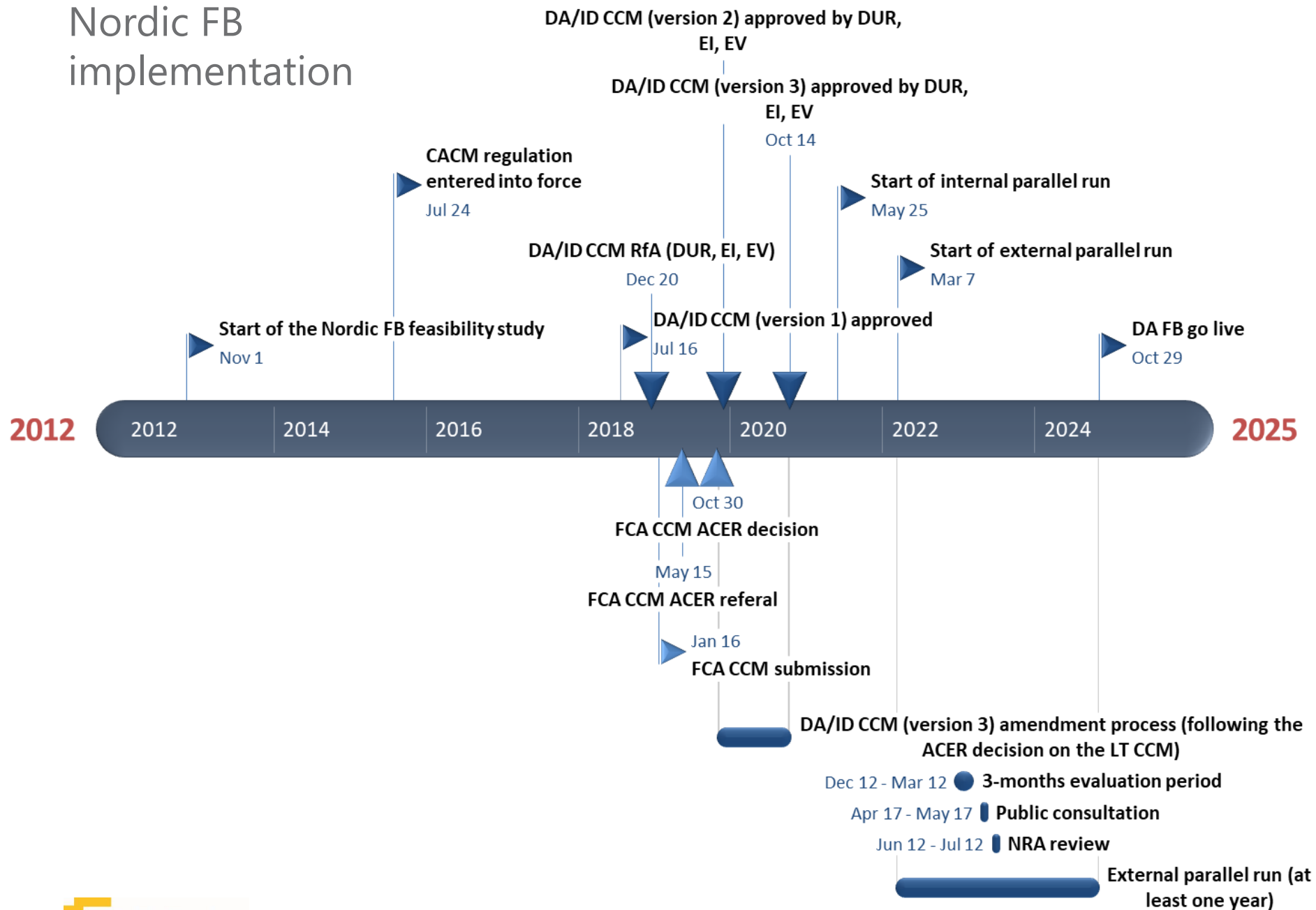
• **Live as of October 29, 2024**

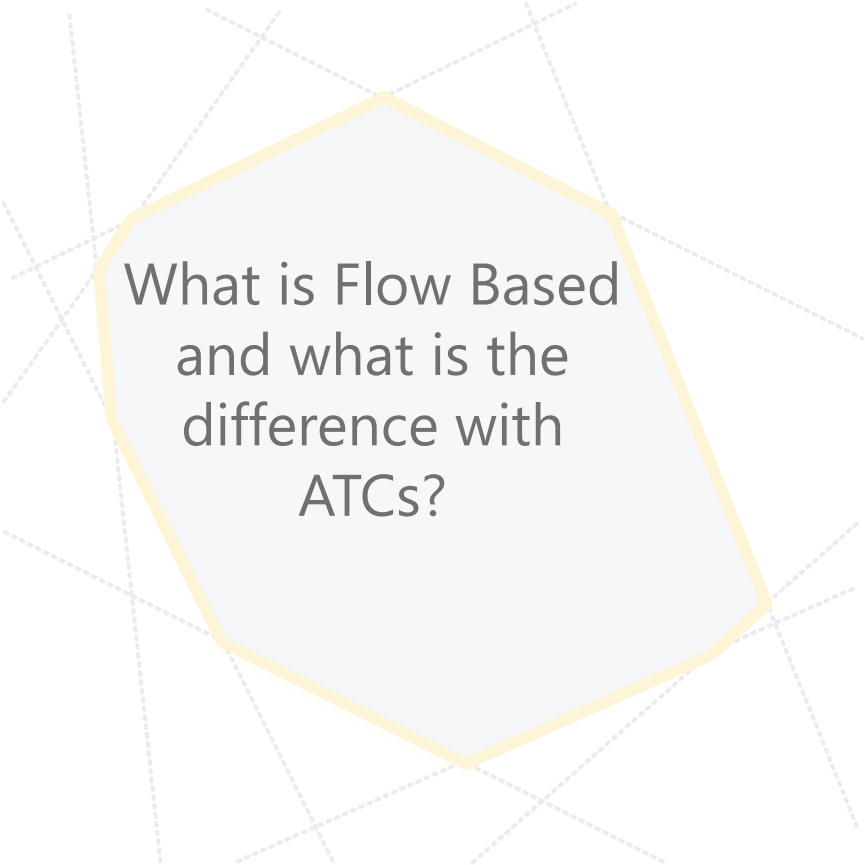
CWE: FB MC

• **Live as of May 21, 2015**

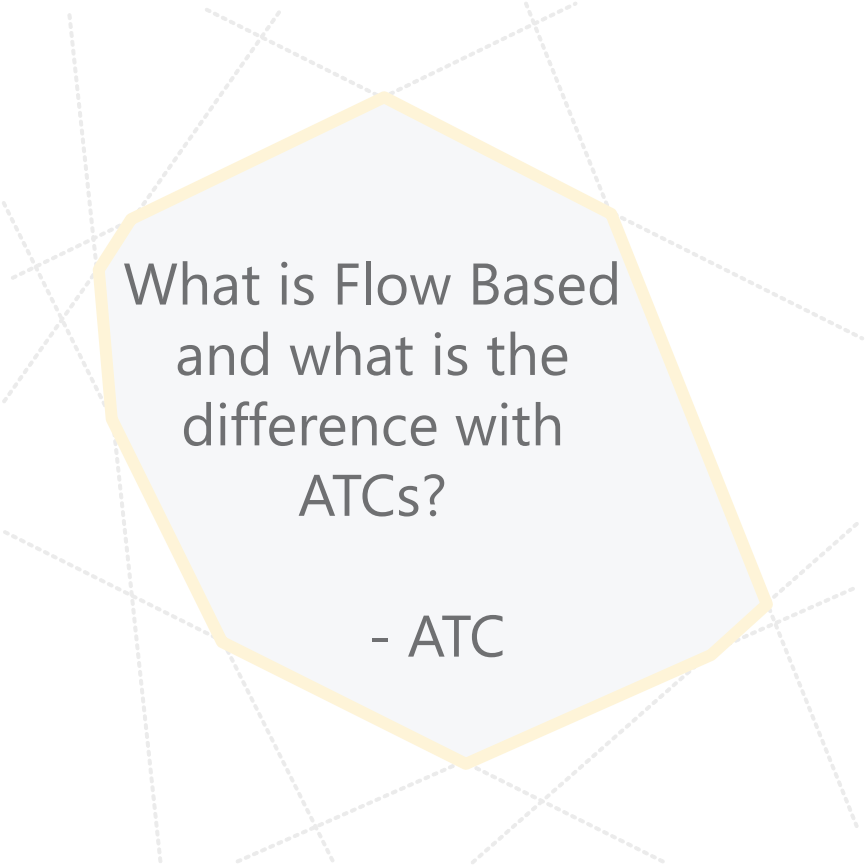


Nordic FB implementation





What is Flow Based
and what is the
difference with
ATCs?

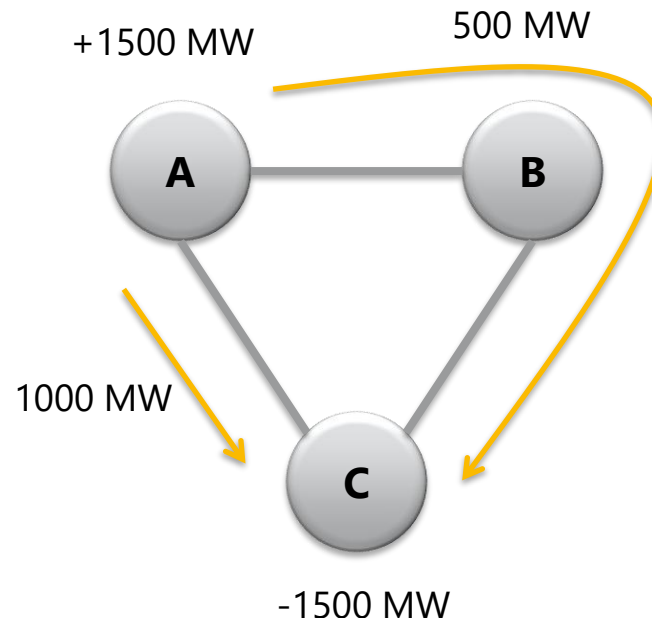
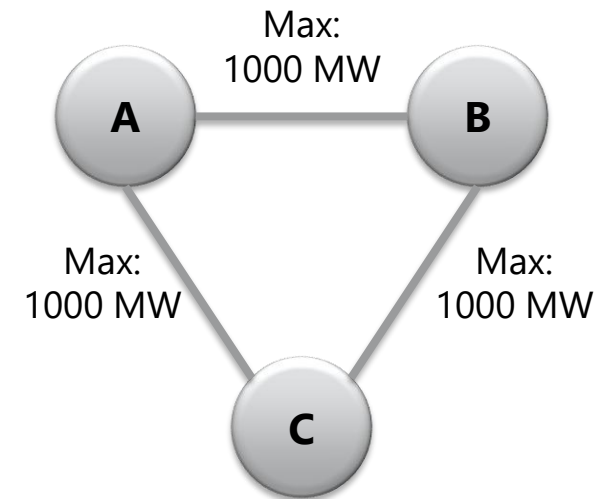


What is Flow Based
and what is the
difference with
ATCs?

- ATC

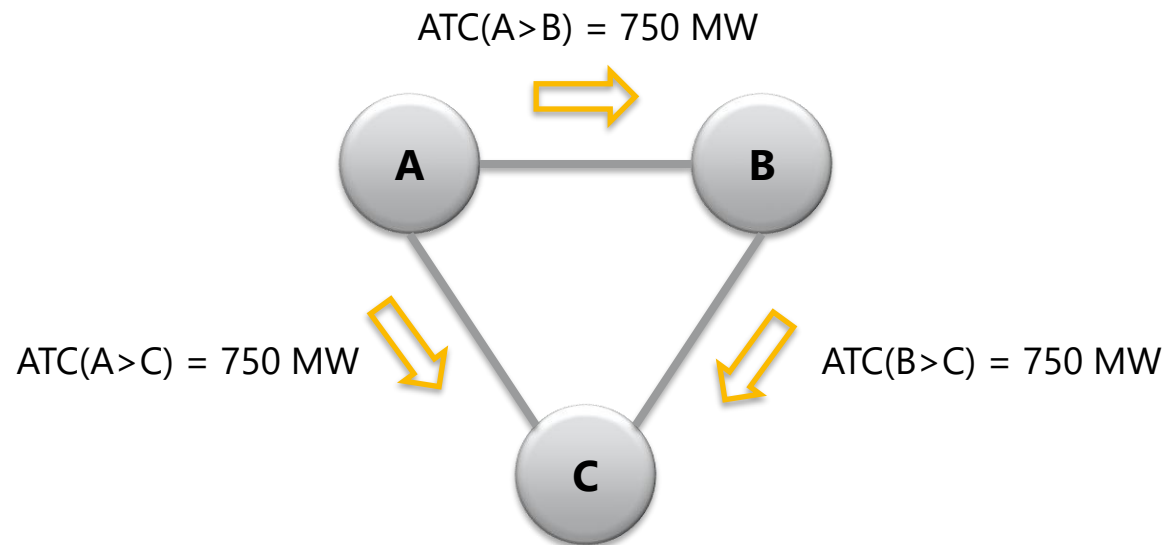
An example three-node network

- Let's consider a three-node network
 - Equal impedances
 - Max flow on the branches: 1000 MW
- The maximum export from A to another bidding area amounts 1500 MW:



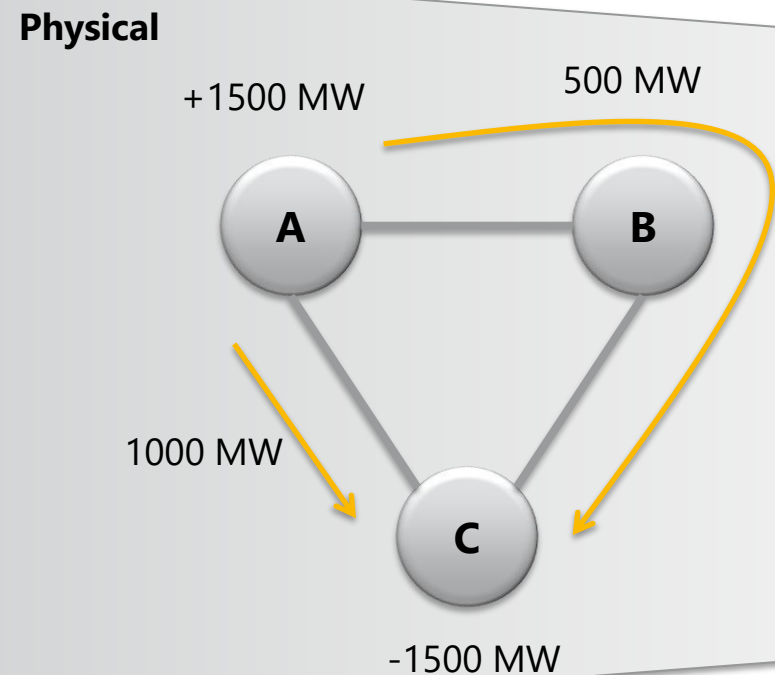
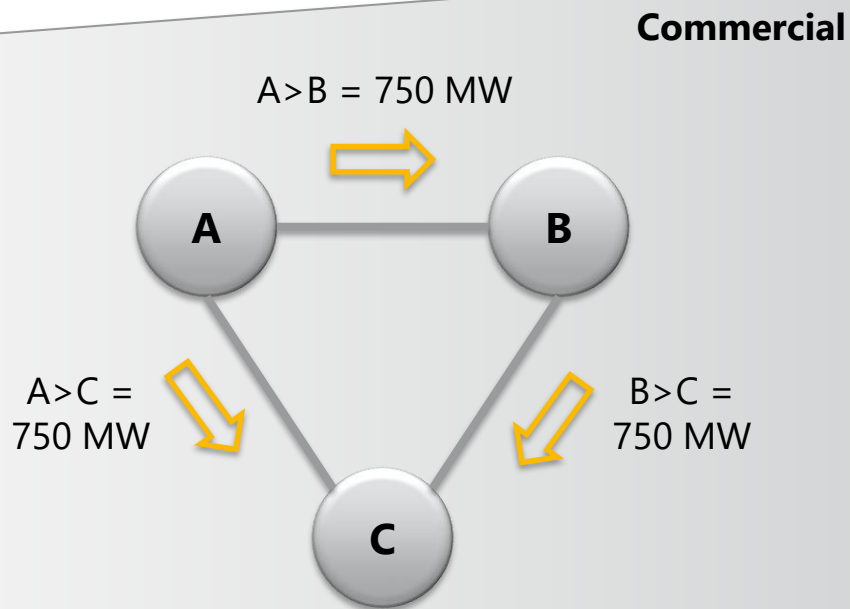
An example three-node network: ATCs

- ATCs are determined by the TSOs to facilitate the market while safeguarding the grid
 - An ATC limits a commercial exchange between two bidding areas
 - ATCs are simultaneously feasible
- Given the maximum export of bidding area A, the TSO needs to split the 1500 MW export capability into two bilateral exchanges, for example:
 - $ATC(A>B) = 750 \text{ MW}$
 - $ATC(A>C) = 750 \text{ MW}$



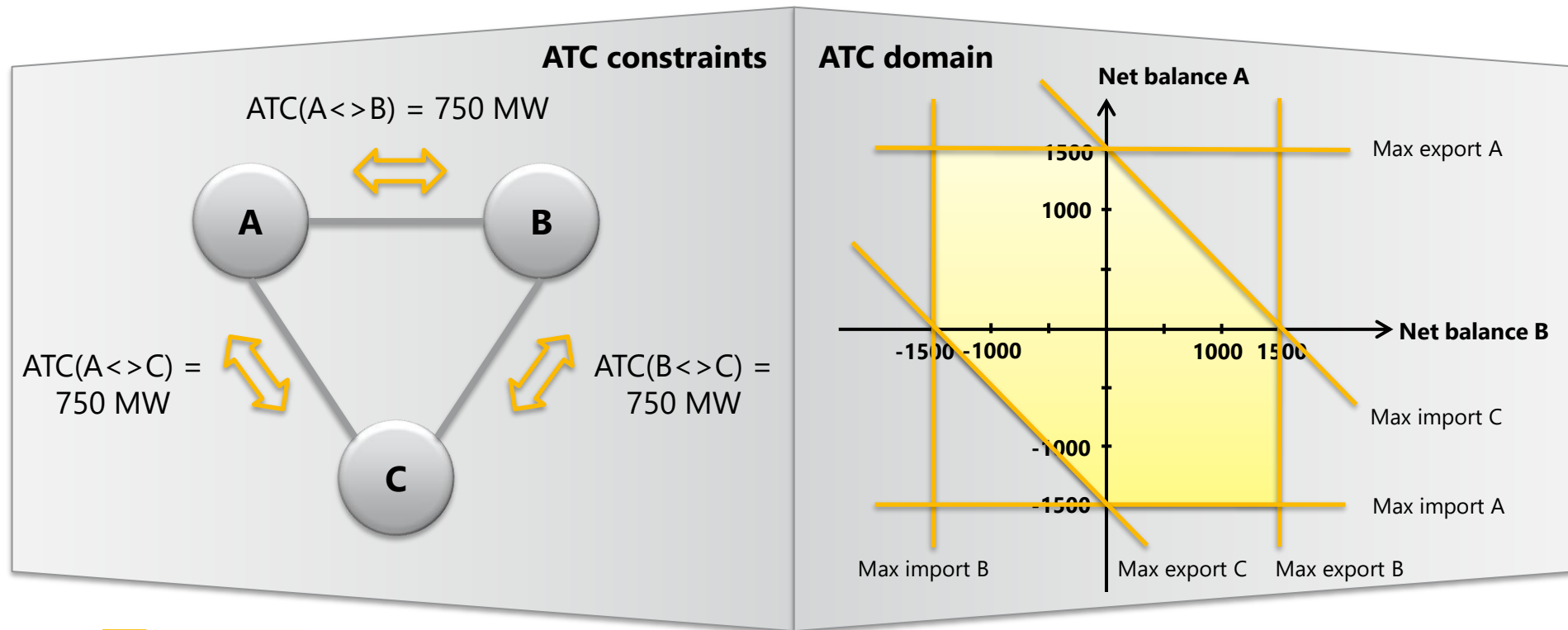
An example three-node network: ATCs and physical flows

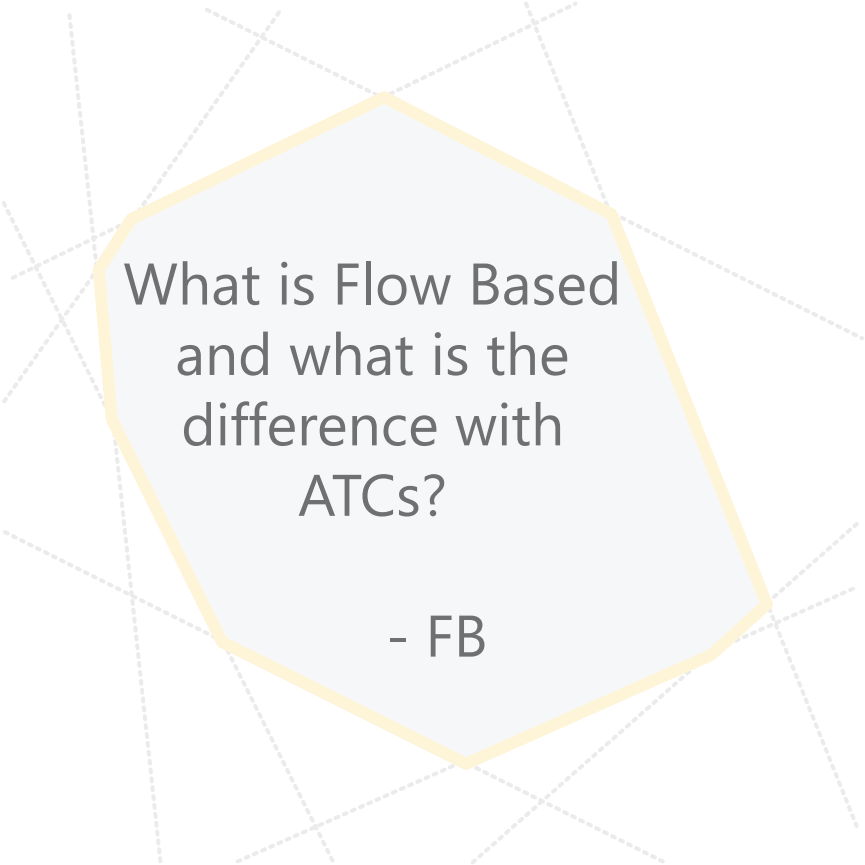
- The following commercial exchanges are feasible given the ATCs:
 - $A > C = 750 \text{ MW}$
 - $A > B = 750 \text{ MW}$
 - $B > C = 750 \text{ MW}$



An example three-node network: ATC domain

- The ATCs in the three-node system define the ATC domain: the import/export positions that the market is allowed to reach under the market coupling while not jeopardizing the grid security



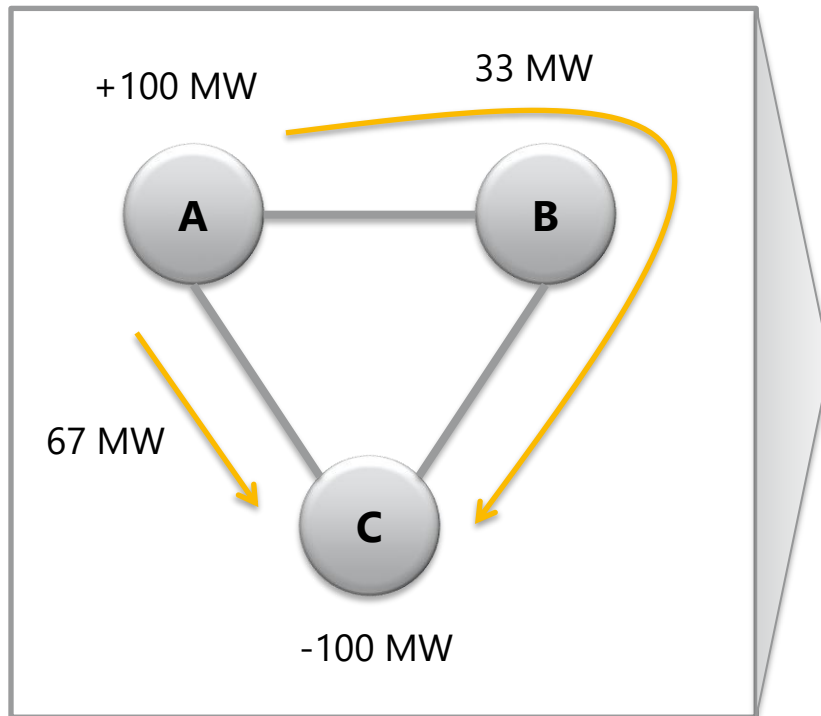


What is Flow Based
and what is the
difference with
ATCs?

- FB

An example three-node network: FB constraints

- FB constraints are a kind of simplified grid model, reflecting the impact of import/export positions on the flows on the grid

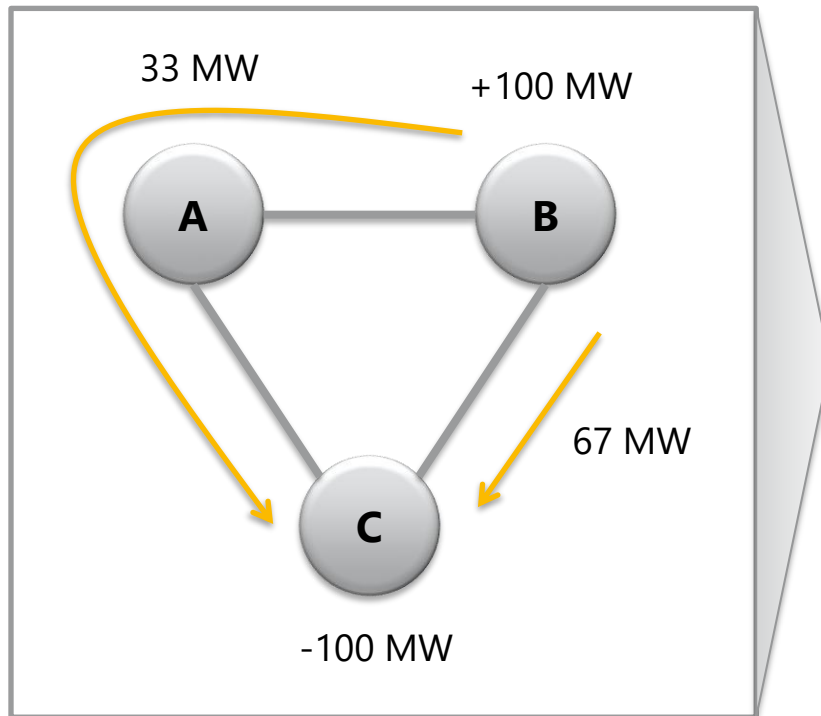


FB constraints ('grid model'):

Line	Margins	PTDF factors		
	Maximum flow	Influence from area A	Influence from area B	Influence from area C
A>B	1000 MW	33 %		
B>C	1000 MW	33 %		
A>C	1000 MW	67 %		

An example three-node network: FB constraints

- FB constraints are a kind of simplified grid model, reflecting the impact of import/export positions on the flows on the grid

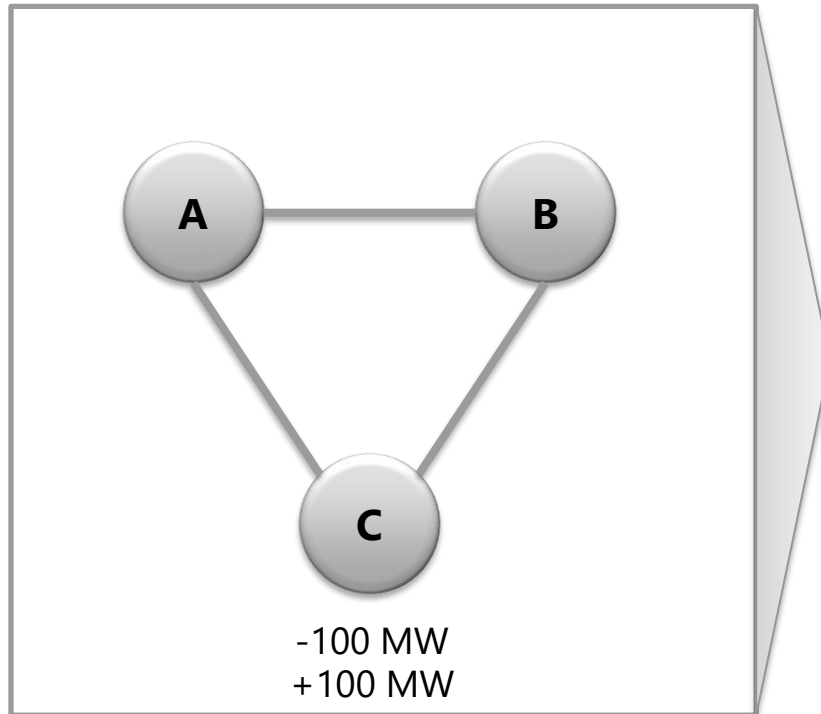


FB constraints ('grid model'):

Line	Margins	PTDF factors		
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A>B	1000 MW	33 %	- 33 %	
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An example three-node network: FB constraints

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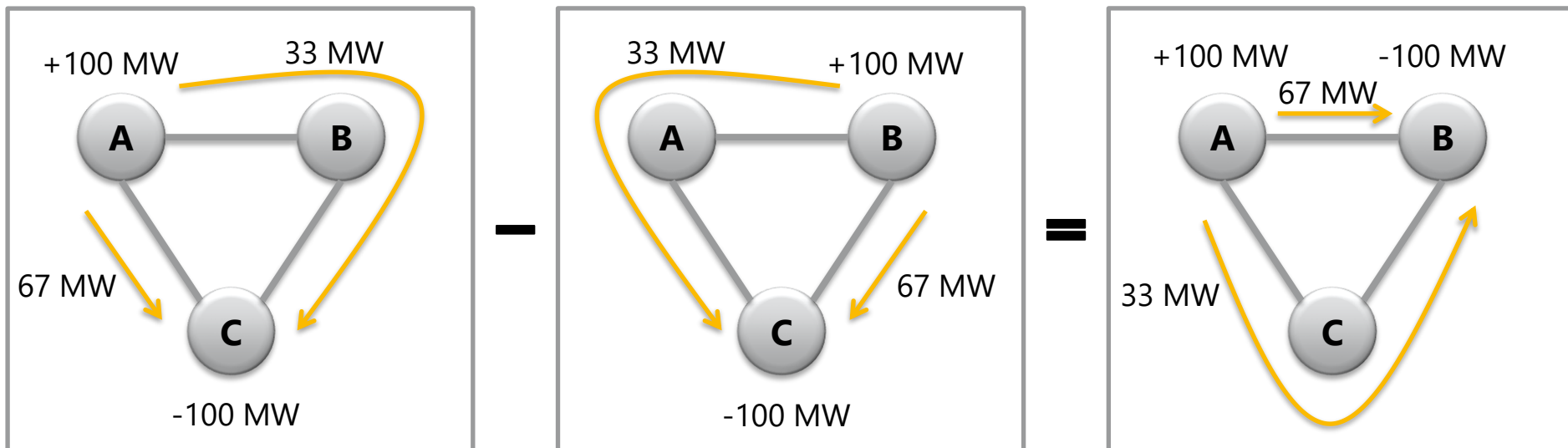


FB constraints ('grid model'):

Line	Margins	PTDF factors		
	Maximum flow	Influence from area A	Influence from area B	Influence from area C
A>B	1000 MW	33 %	- 33 %	0
B>C	1000 MW	33 %	67 %	0
A>C	1000 MW	67 %	33 %	0

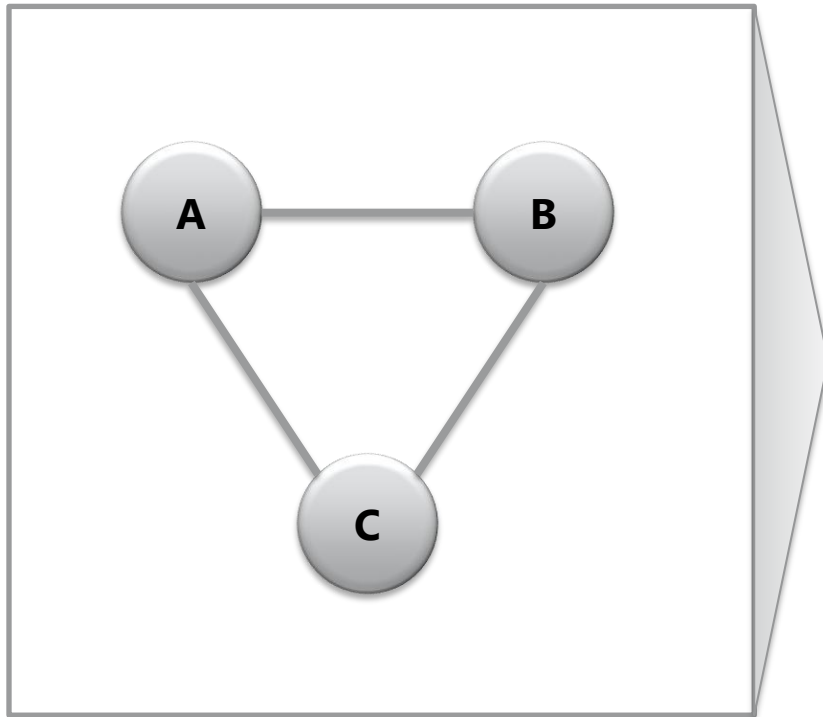
An example three-node network: FB and the reference node

- An exchange of 100 MW from bidding area A to bidding area B is equivalent to: an exchange of 100 MW from area A to the reference node C – an exchange of 100 MW from area B to the reference node C
- This property holds due to the linearity of the PTDF computation (DC load flow)



An example three-node network: FB constraints

- FB constraints are a kind of simplified grid model, reflecting the impact of import/export positions on the flows on the grid



FB constraints ('grid model'):

Line	Margins	PTDF factors		
	Maximum flow	Influence from area A	Influence from area B	Influence from area C
A>B	1000 MW	33 %	- 33 %	0
B>C	1000 MW	33 %	67 %	0
A>C	1000 MW	67 %	33 %	0
B>A	1000 MW	-33 %	33 %	0
C>B	1000 MW	- 33 %	- 67 %	0
C>A	1000 MW	- 67 %	- 33 %	0

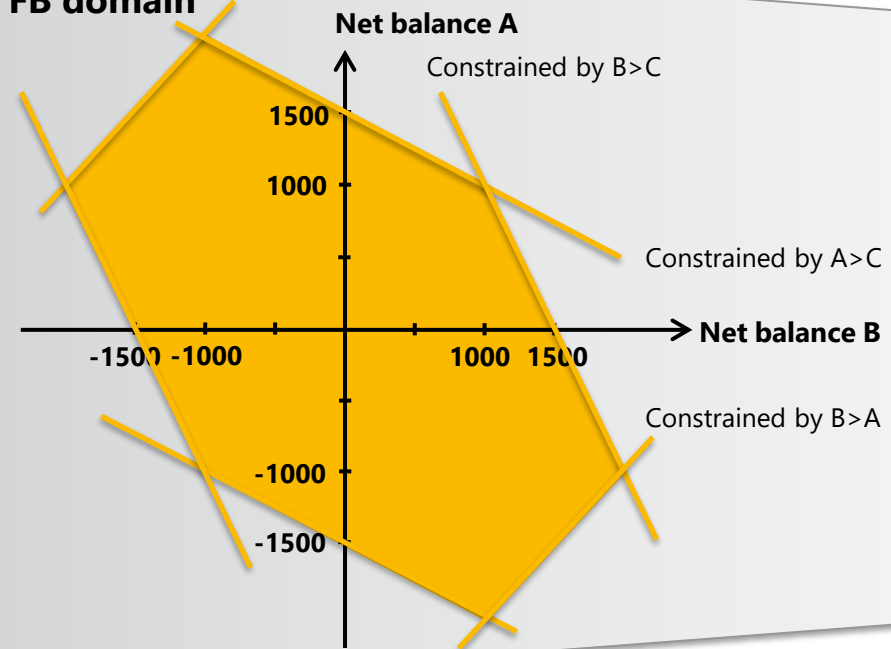
An example three-node network: FB domain

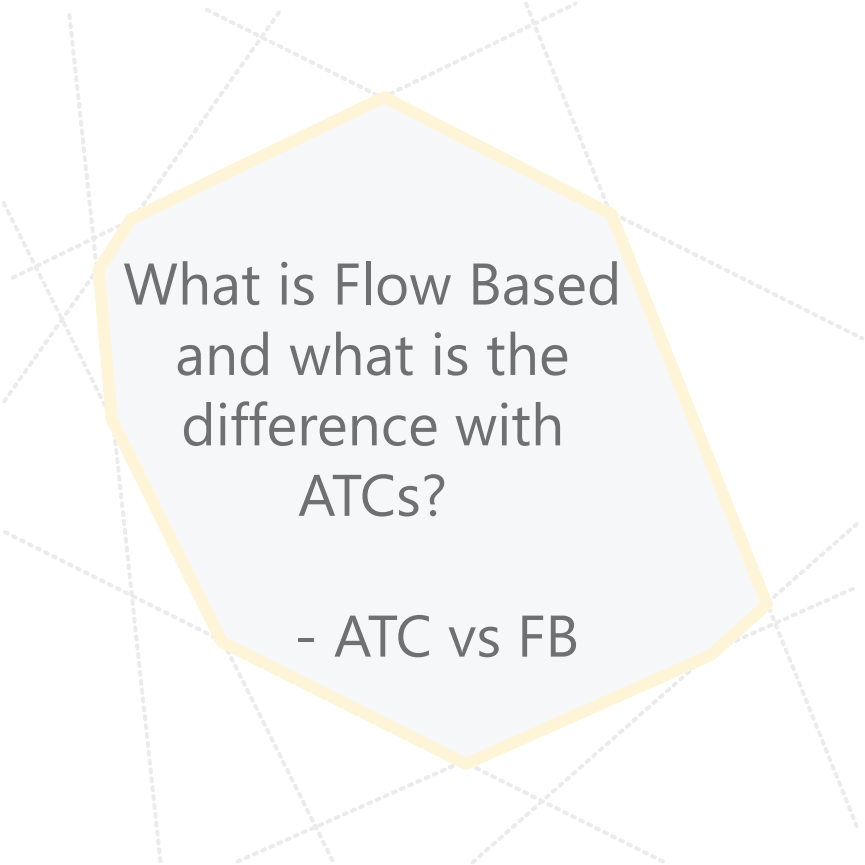
- The FB constraints in the three-node system define the FB domain: the import/export positions that the market is allowed to reach under the market coupling while not jeopardizing the grid security

FB constraints

Line	Maximum flow	Influence from area A	Influence from area B	Influence from area C
A > B	1000 MW	33 %	- 33 %	0
B > C	1000 MW	33 %	67 %	0
A > C	1000 MW	67 %	33 %	0
B > A	1000 MW	- 33 %	33 %	0
C > B	1000 MW	- 33 %	- 67 %	0
C > A	1000 MW	- 67 %	- 33 %	0

FB domain



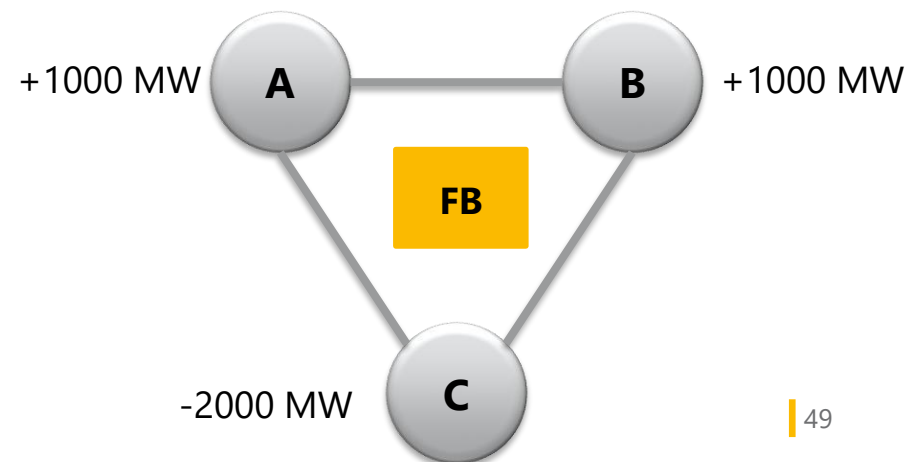
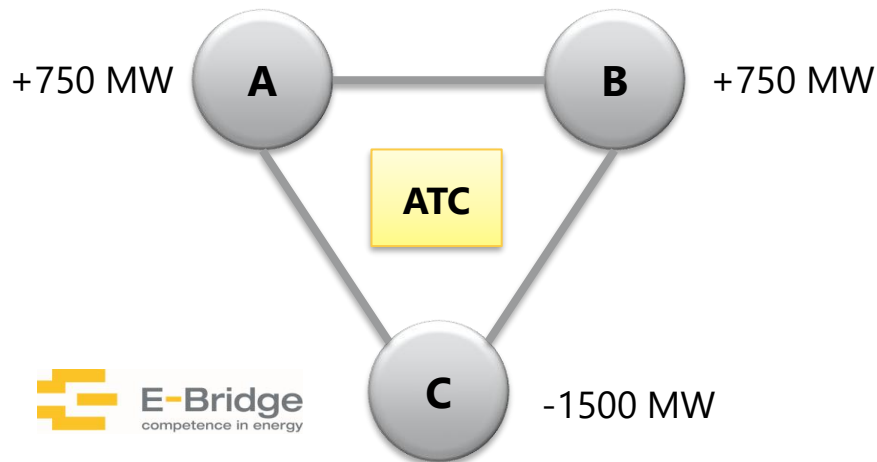
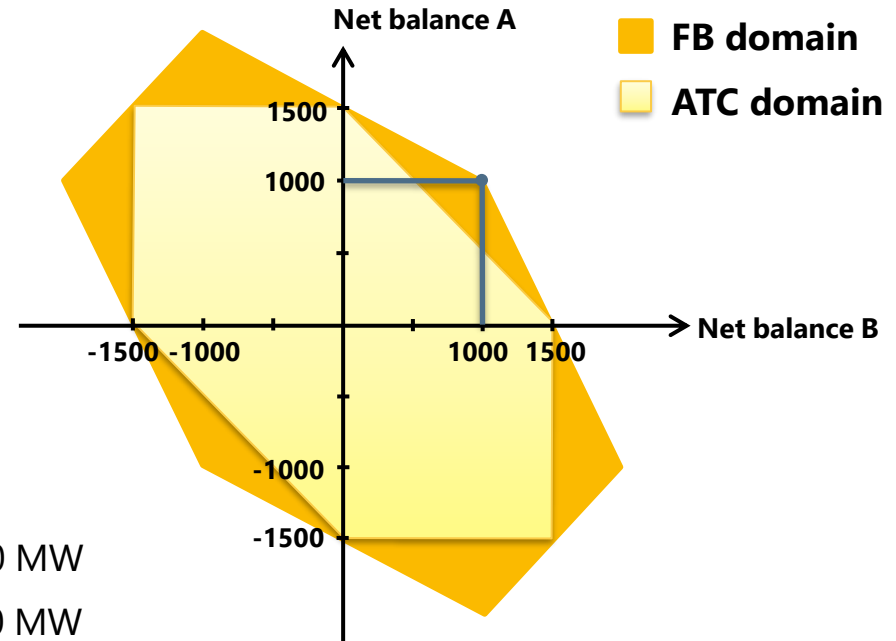


What is Flow Based
and what is the
difference with
ATCs?

- ATC vs FB

An example three-node network: ATC vs FB domain

- In FB capacity split is not a choice of the TSO, but is market driven (at the time of allocation)
- FB offers more trading opportunities with the same level of security of supply
- Example:
 - ATC: North-South exchange limited to 1500 MW
 - FB: North-South exchange possible of 2000 MW

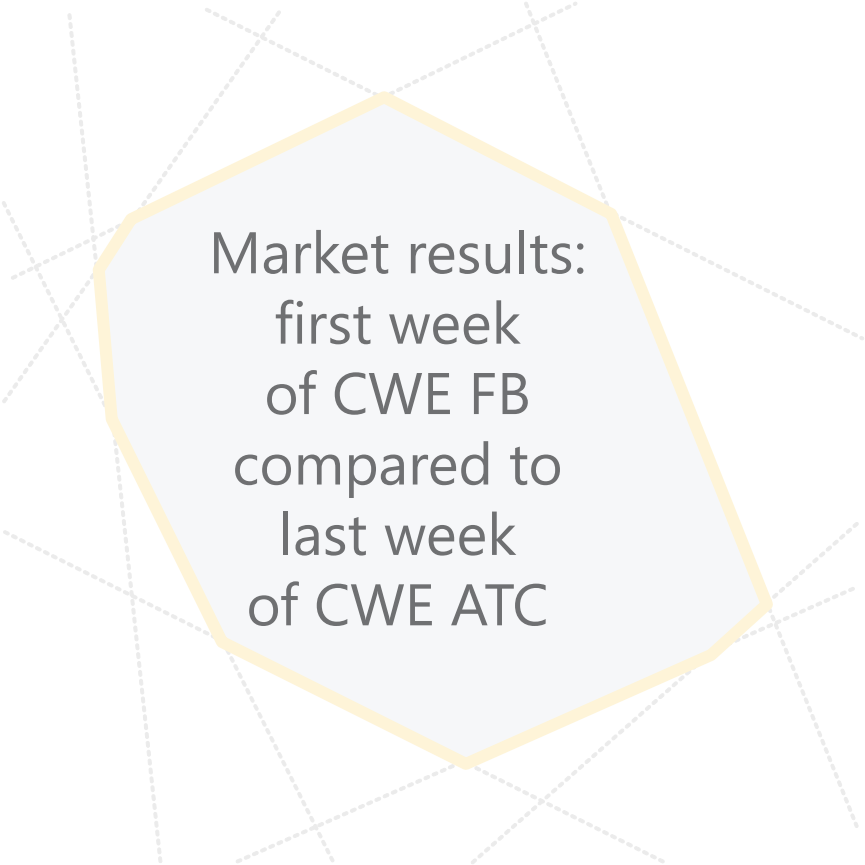




Advantages of the FB approach

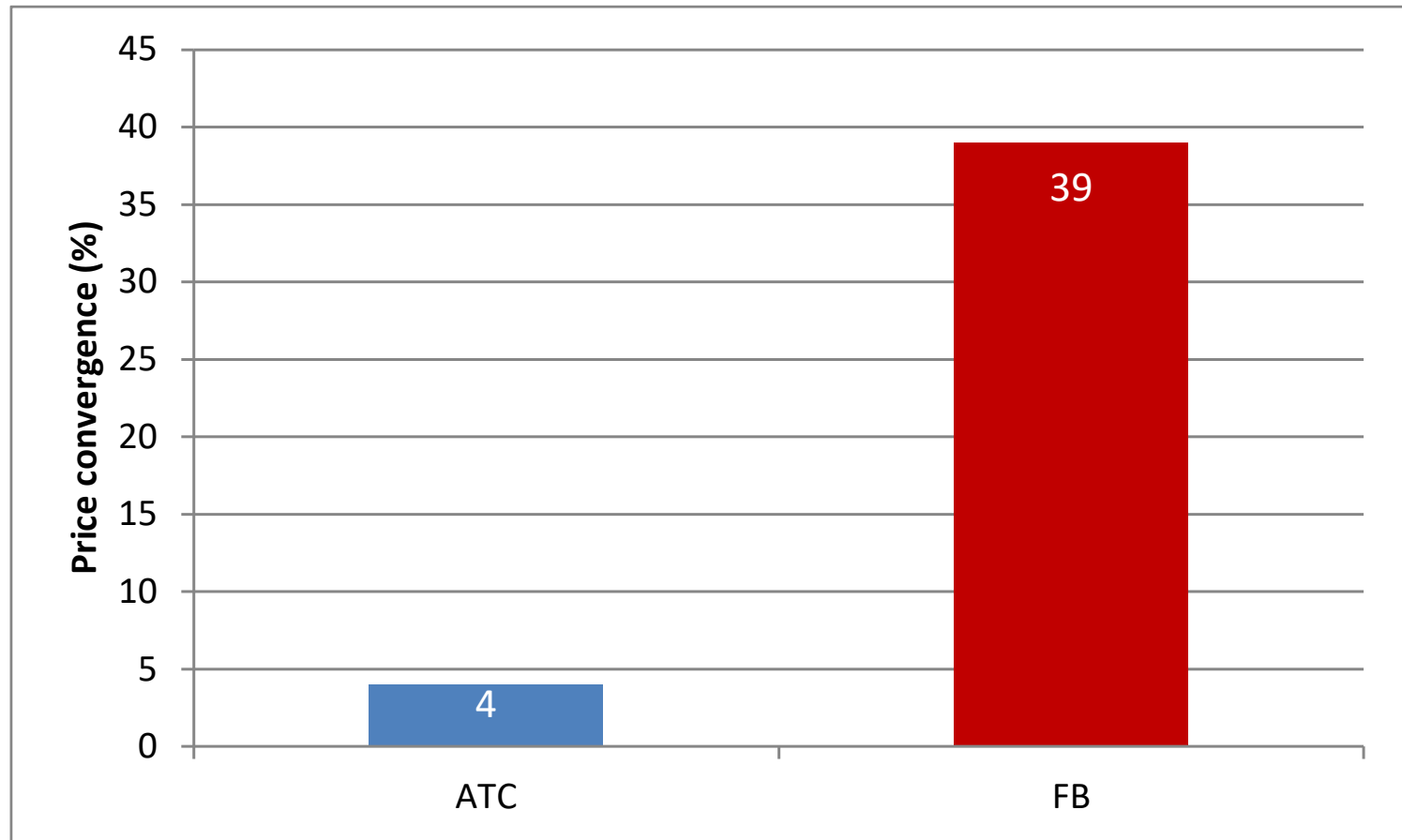
Advantages of the FB approach

- In FB capacity split is not a choice of the TSO, but is market driven (at the time of allocation)
 - More efficient and flexible use of the grid
 - FB offers more trading opportunities with the same level of security of supply
 - More price convergence / smaller price differences
 - Higher social welfare
 - Income redistribution
 - FB offers the possibility to have the DC cables efficiently embedded in the allocation mechanism, by providing a fair competition for the use of the scarce AC capacity
- Flow-based market coupling provides an efficient allocation mechanism in which all exchanges that are subject to the allocation mechanism compete with one another for the use of the scarce capacity

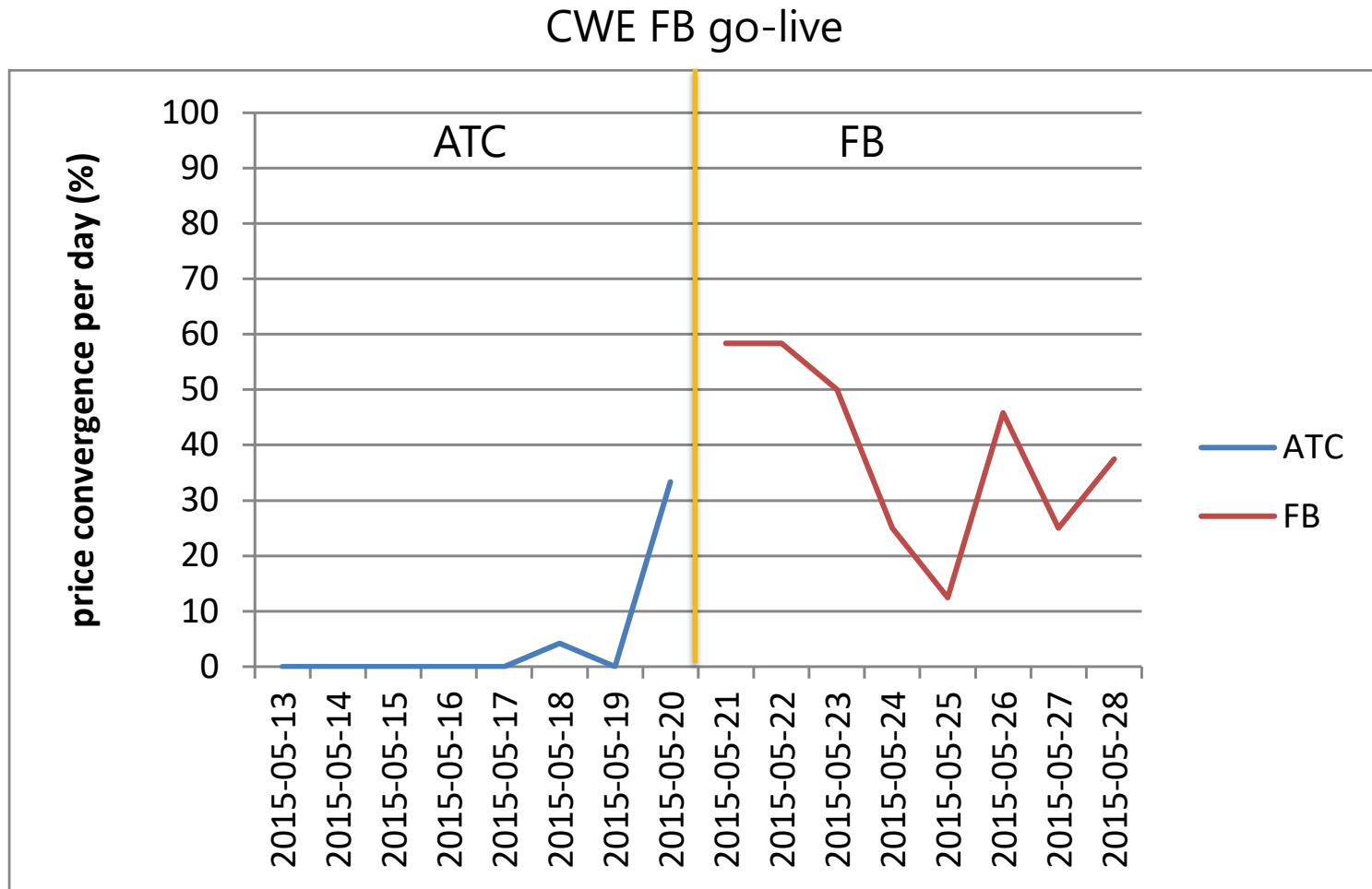


Market results:
first week
of CWE FB
compared to
last week
of CWE ATC

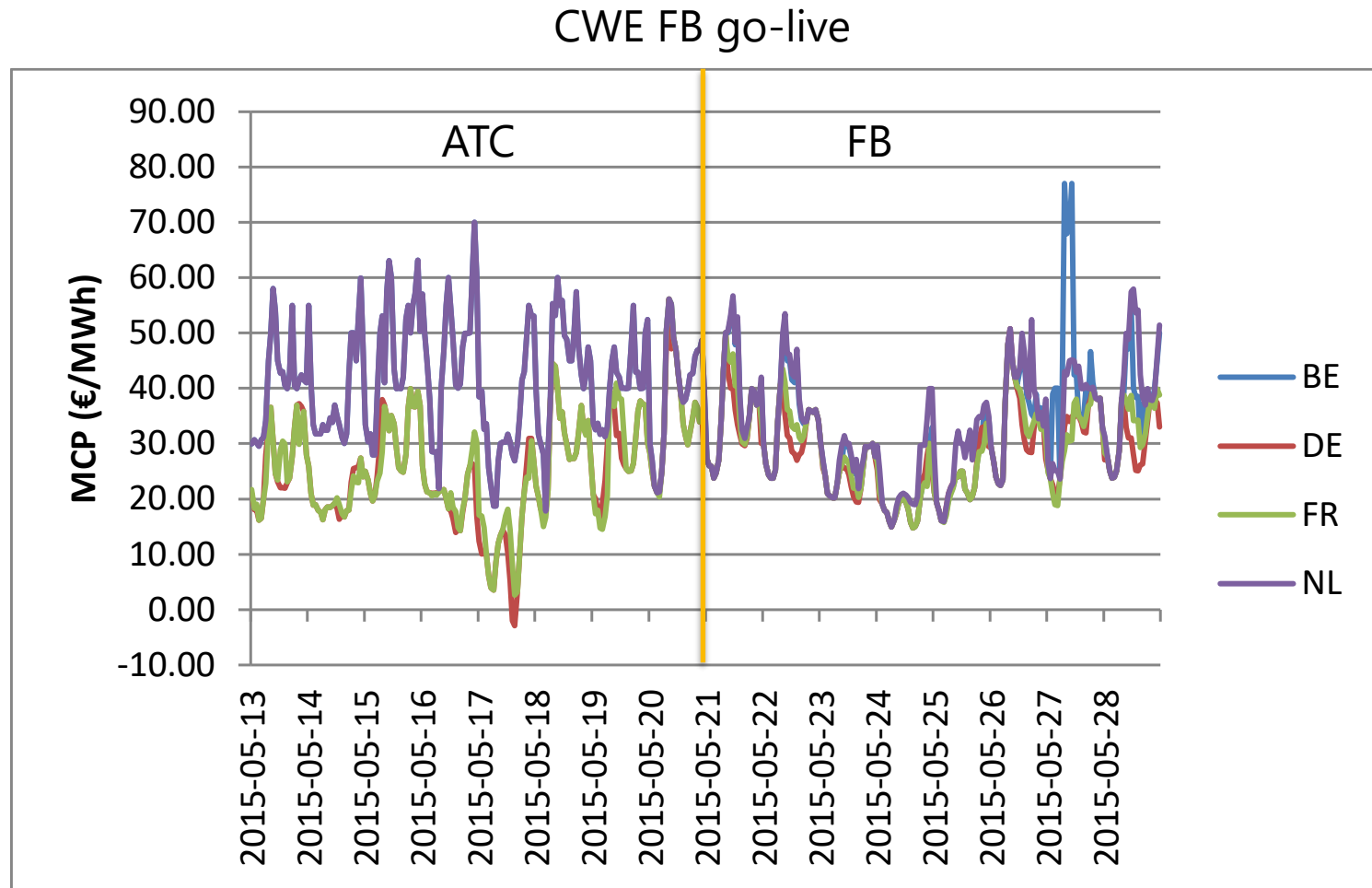
Average price convergence – 8 days before and after CWE FB go-live



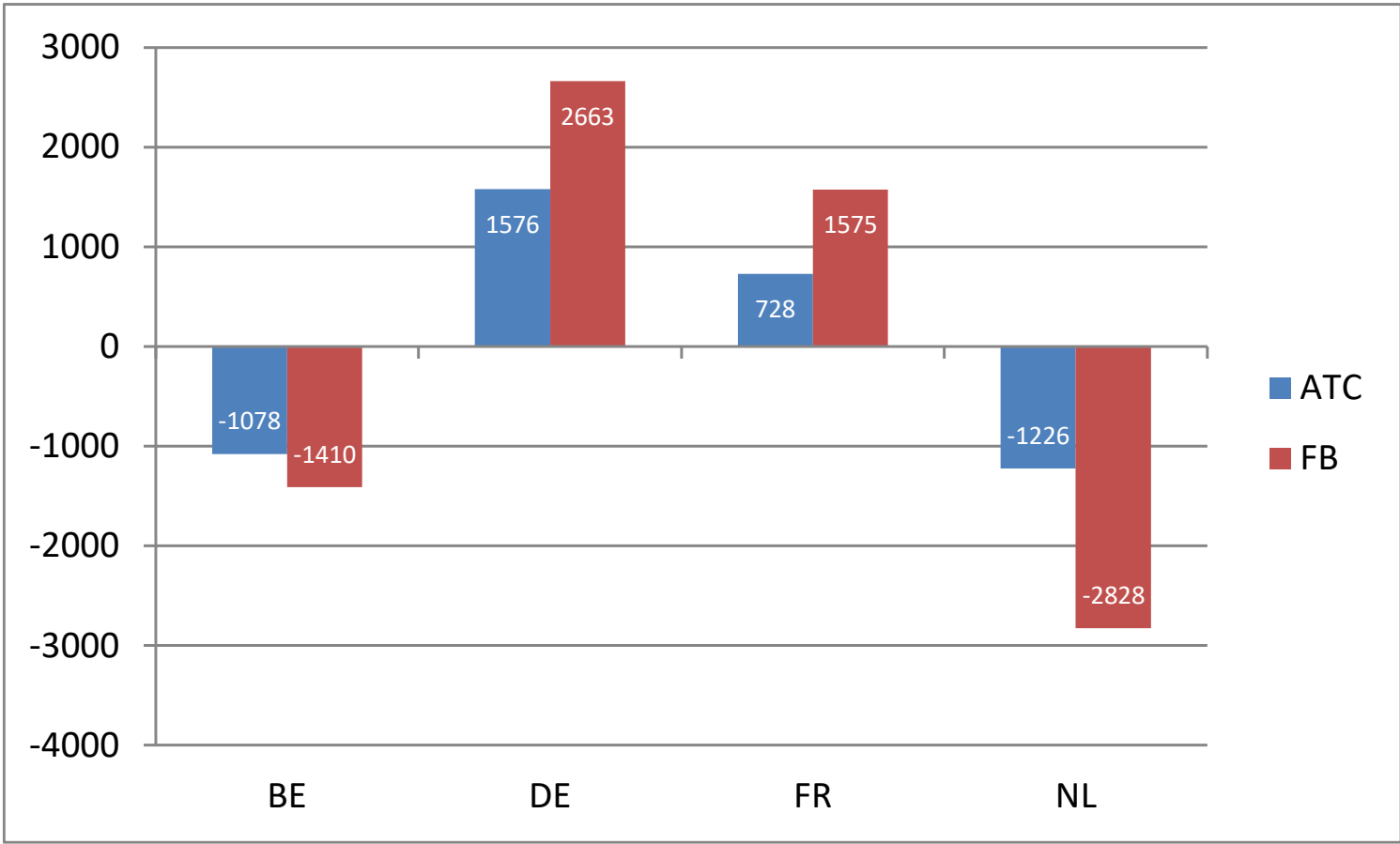
Daily price convergence – 8 days before and after CWE FB go-live



Market clearing prices – 8 days before and after CWE FB go-live



Average CWE MC net positions – 8 days before and after CWE FB go-live





FB allocation and price formation

Prices under a FBMC

- FBMC optimization

- Objective function: Maximize social welfare
- Control variables: Net positions
- Subject to: $\sum \text{net positions} = 0$

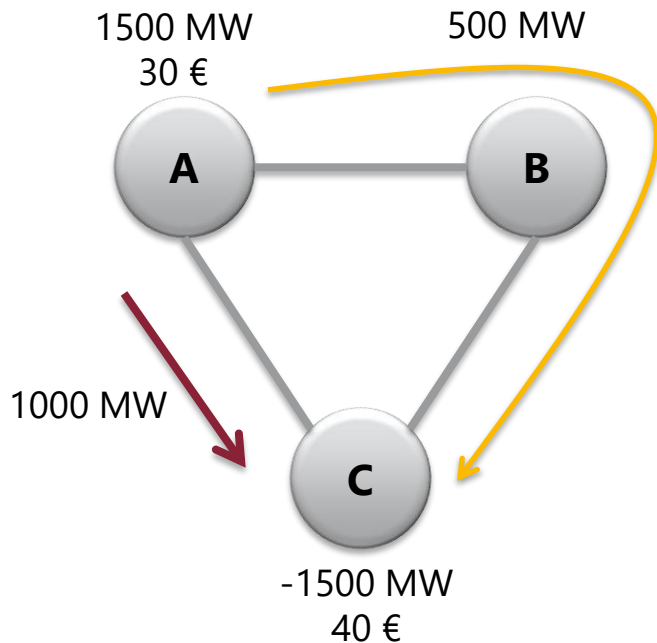
Grid constraints

- In case of congestion, the grid constraint receives a shadow price (μ): the increase of the objective function (being the social welfare) when the constraint is relieved with 1 MW

- Price relation under FB:

$$MCP_i - MCP_j = \sum_{cb} (PTDF_j^{cb} - PTDF_i^{cb}) \cdot \mu_{cb}$$

Prices under a FBMC: an example



Line	Maximum flow	Influence from area A	Influence from area B	Influence from area C
A>B	1000 MW	33 %	- 33 %	0
B>C	1000 MW	33 %	67 %	0
A>C	1000 MW	67 %	33 %	0
B>A	1000 MW	-33 %	33 %	0
C>B	1000 MW	- 33 %	- 67 %	0
C>A	1000 MW	- 67 %	- 33 %	0

- Price relation under FB:

$$MCP_C - MCP_A = (PTDF_A - PTDF_C) \cdot \mu$$

- Shadow price equals $\mu = 15 \text{ €/MW}$

$$40 - 30 = (0.67 - 0) \cdot 15$$

- Price in area B:

$$MCP_B - MCP_A = (PTDF_A - PTDF_B) \cdot \mu$$

$$MCP_B - 30 = (0.67 - 0.33) \cdot 15$$

$$MCP_B = 35$$

Prices under a FBMC: consequences

- In case there is no congestion, all bidding zones have the same price
- In case of congestion, the prices of the bidding zones are set in accordance to their electrical impact on the binding constraint (i.e. the PTDF factors)



Observations from
the first year
with the Nordic
DA FB

Key takeaways post FB go-live

- Operation has been good and secure
- Flow-based has increased capacity and moved it closer to the operational boundary.
 - Flow-based will continue to be adjusted and improved as it is an ongoing process aimed at finding the right balance.
- The overall performance of the FB capacity calculation and allocation in the Nordic area is presented in its first year in operation (November 2024 – October 2025). The results are compared to the NTC results from the period December 2022 – October 2024.

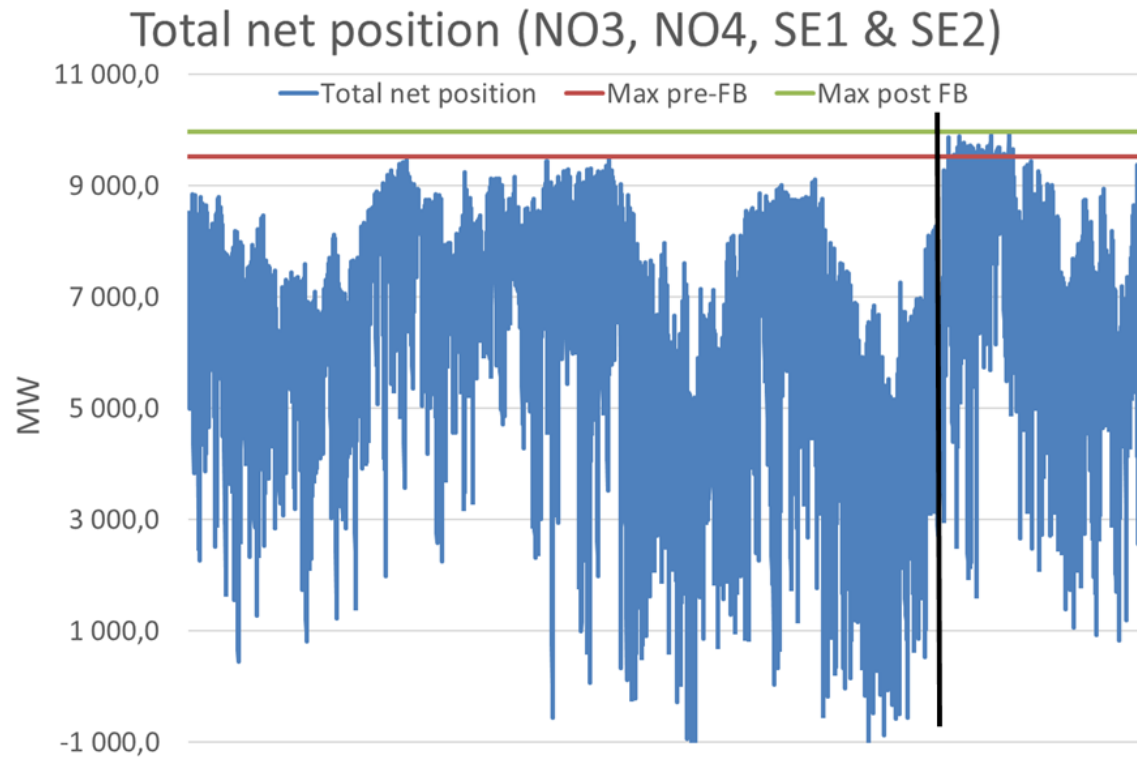
Changes in flow

- North Cut:
Average flow is increased by 32%
Maximum flow decreased by 6% 4320
➔ 4074 MWh
- Central Cut:
Average flow is increased by 31%
Maximum flow increased by 9% 8023 ➔
8711 MWh
- South Cut:
Average flow is increased by 0,2%
Maximum flow increased by 9% 5336 ➔
5790 MWh
- South Norway:
Average flow is increased by 243%
Maximum flow increased by 23% 2564
➔ 3156 MWh
- External cables:
Average flow is increased by 6,32%
Maximum flow similar 9103 ➔
9106 MWh



Net position in the Northern bidding zones

- Total net position in NO3, NO4, SE1 & SE2 before and after go-live (January 2021-November 2025)
- The maximum export from the Northern bidding zones (i.e. NO3, NO4, SE1, and SE2) has increased by 440 MW
- The maximum export observed in the more than 3,5 years before FB go-live was exceeded already in 287 hours in the one year after the go-live



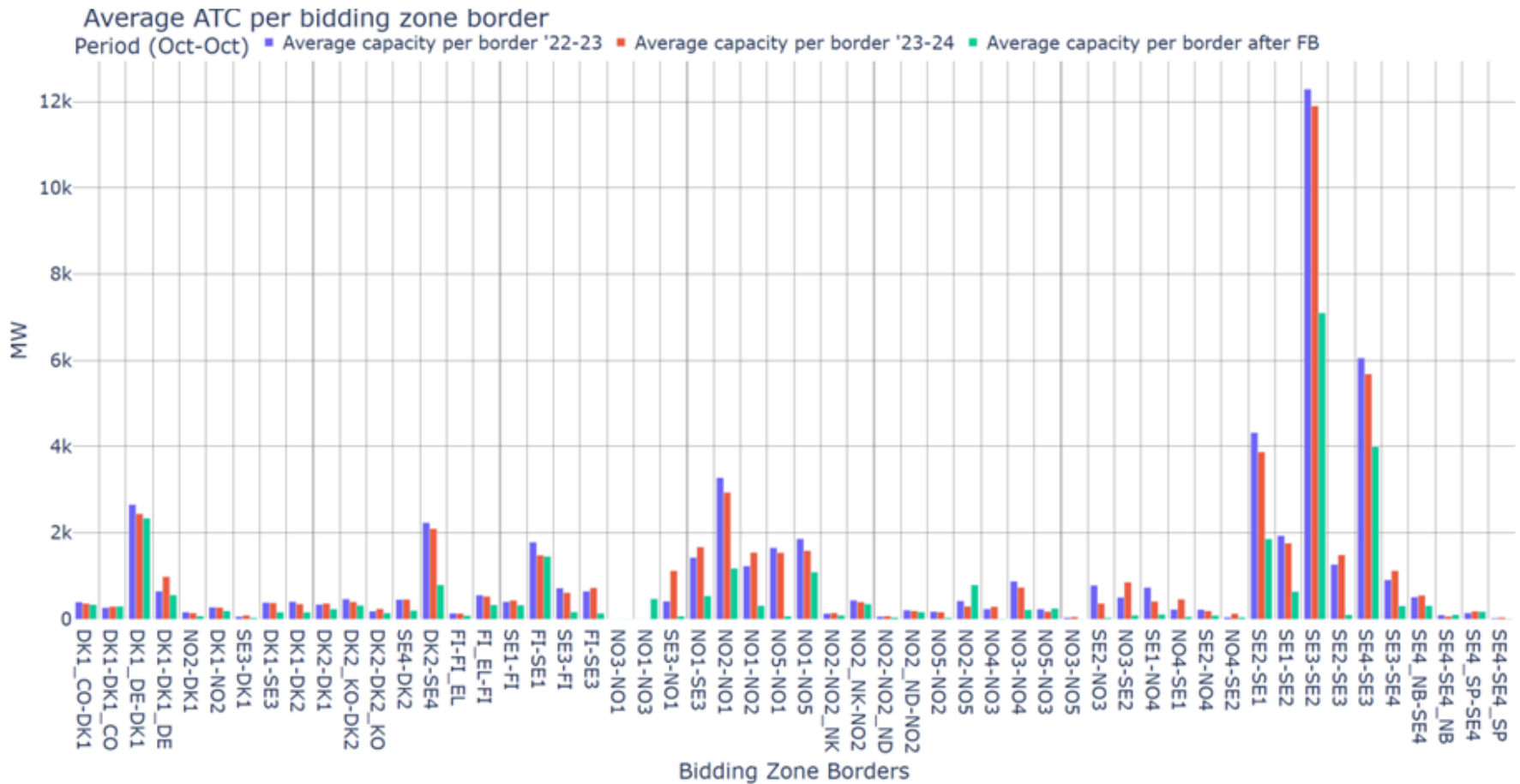
Average prices

- Average bidding zone prices in the one year before FB go-live and the one year after, are tabulated
- Average price difference between the Northern and Southern bidding zones has increased since the introduction of FB
 - In the Northern bidding zones (SE1, SE2, NO3, and NO4), the prices have decreased compared to the year before the FB go-live. This is mainly driven by high wind production and high water reservoir levels
 - In the Southern bidding zones, including SE3, NO1 and NO5, the prices have increased compared to the year before the FB go-live. The increase can most likely be attributed to the following factors: higher prices in Germany, low nuclear production, and low water reservoir levels in Southern Norway

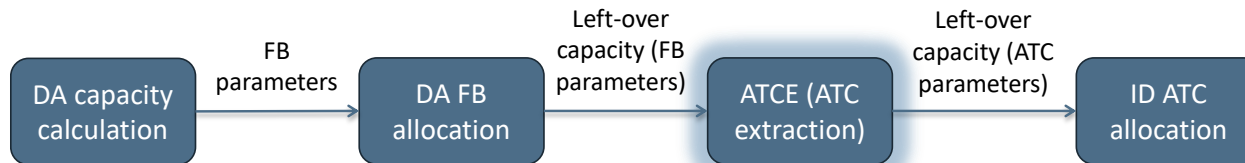


Bidding zone	Average price per bidding zone (EUR)		
	1.nov 23 - 31 okt 24	1.nov 24 - 31 okt 25	Difference
NO1	47,7	54,6	6,9
NO2	54,2	62,8	8,7
NO3	35,7	15,9	-19,9
NO4	31,4	5,2	-26,2
NO5	48,1	42,2	-5,9
SE1	32,2	14,2	-18,0
SE2	32,3	13,6	-18,8
SE3	38,6	46,0	7,5
SE4	50,7	60,5	9,8
FI	50,7	40,5	-10,3
DK1	68,1	82,2	14,0
DK2	67,9	83,5	15,7
GER	73,3	91,6	18,3

Average ID gate opening capacity (i.e. DA left-over capacity)

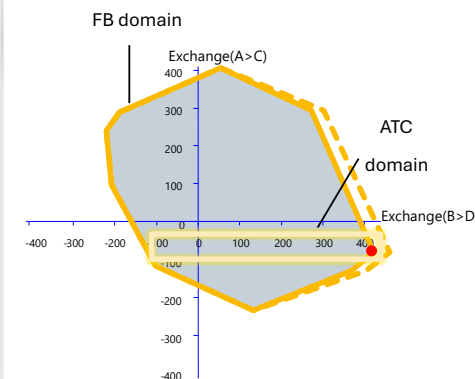
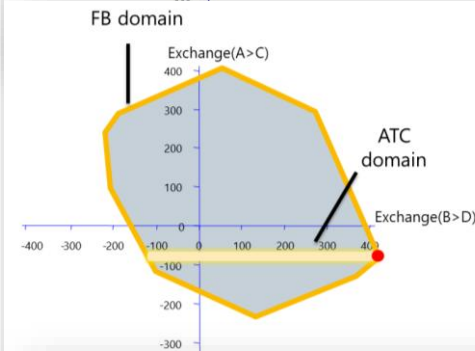
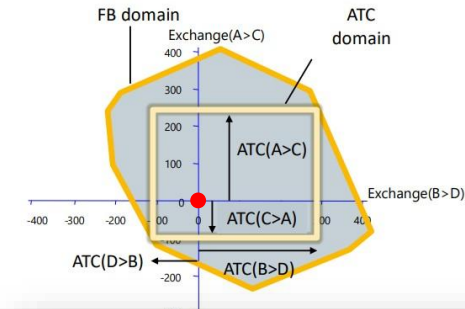


Challenge with ID capacity and ATCE



- The transition from NTC to FB follows a stepwise implementation as it cannot be realized in one big bang for all market timeframes, and this goes along with some inefficiencies.
- One being that ATC is a rudimentary translation from physics to market capacities, and the translation from FB to ATC needs to cope with and respect those shortcomings.
- As a result, we can observe that the **order of magnitude of the ID capacities after the DA FB go live are much lower** than what we observed in the same period before the DA FB go-live.

● Market clearing point



Next steps

- In October 2025, the **year-ahead** FB capacity calculation went live in the Nordics.
 - The year-ahead FB capacity calculation is based on eight scenarios: Spring, Summer, Autumn, and Winter – both in a peak and off-peak situation. The corresponding FB domains are published. As part of the Nordic Capacity Calculation Region internal borders, only the bidding zone border DK1 – DK2 (via the 'Storebælt' HVDC link) is currently subject to long-term allocation. For this border, the ATCE process is operated until the Single Allocation Platform can support FB parameters.
- The **month-ahead** FB capacity calculation in the Nordics is foreseen to go live mid 2026.
 - Each month-ahead FB capacity calculation is based on two scenarios.
- The project to implement the FB methodology for the **intraday** auctions is ongoing in the Nordic CCR. The planning and implementation of this project is closely connected to the European timeline.
- This is followed by the application of the FB approach in the **balancing** timeframe.

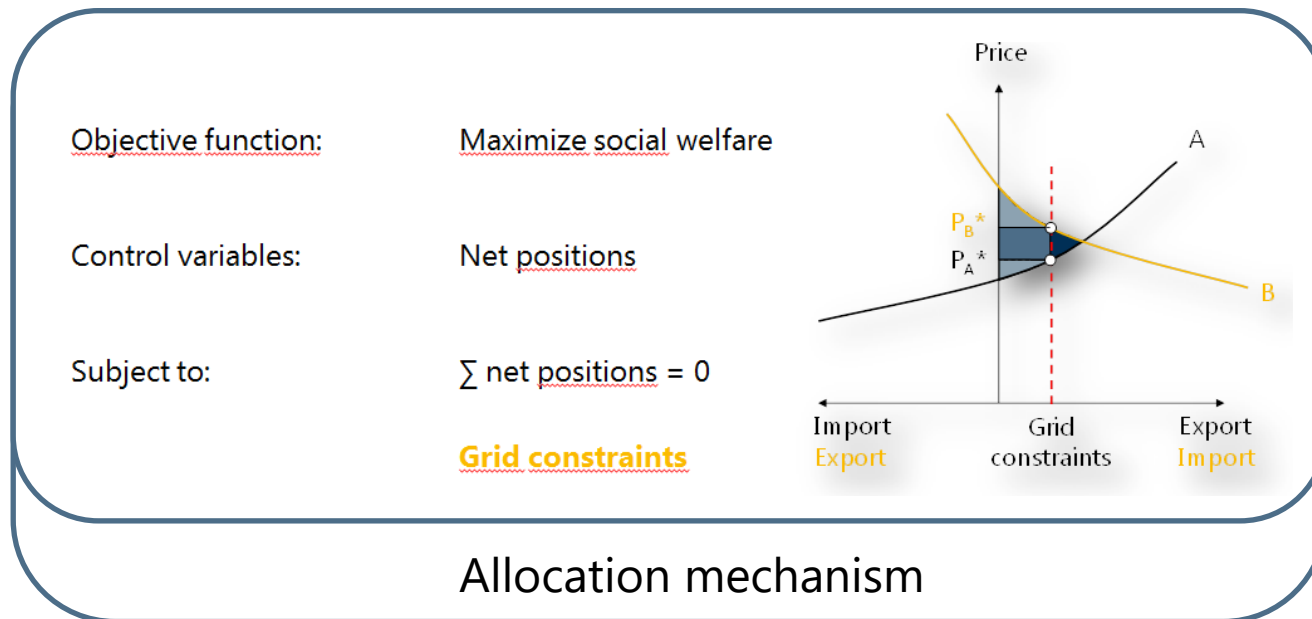


The diagram shows a central light gray polygon with a thick yellow border. This polygon is surrounded by a network of thin, gray dashed lines that intersect at various points, forming a web-like structure. The text "Bidding Zone Delineation" is centered within the polygon.

Bidding Zone Delineation

Allocation mechanism

- In an allocation mechanism, all commercial exchanges (i.e. trades from one market area to another) that are subject to the allocation mechanism are competing for the scarce capacity made available by the TSOs within the allocation mechanism



Zone delimitation

Zone delimitation is one way to manage congestions in the transmission network. Three systems may be distinguished:

- the uniform system, where no exchanges are subject to an allocation mechanism;
- the nodal system, where exchanges between all nodes are subject to an allocation mechanism;
- the zonal system, where only exchanges between zones are subject to an allocation mechanism.



Source: website Nordpool Spot

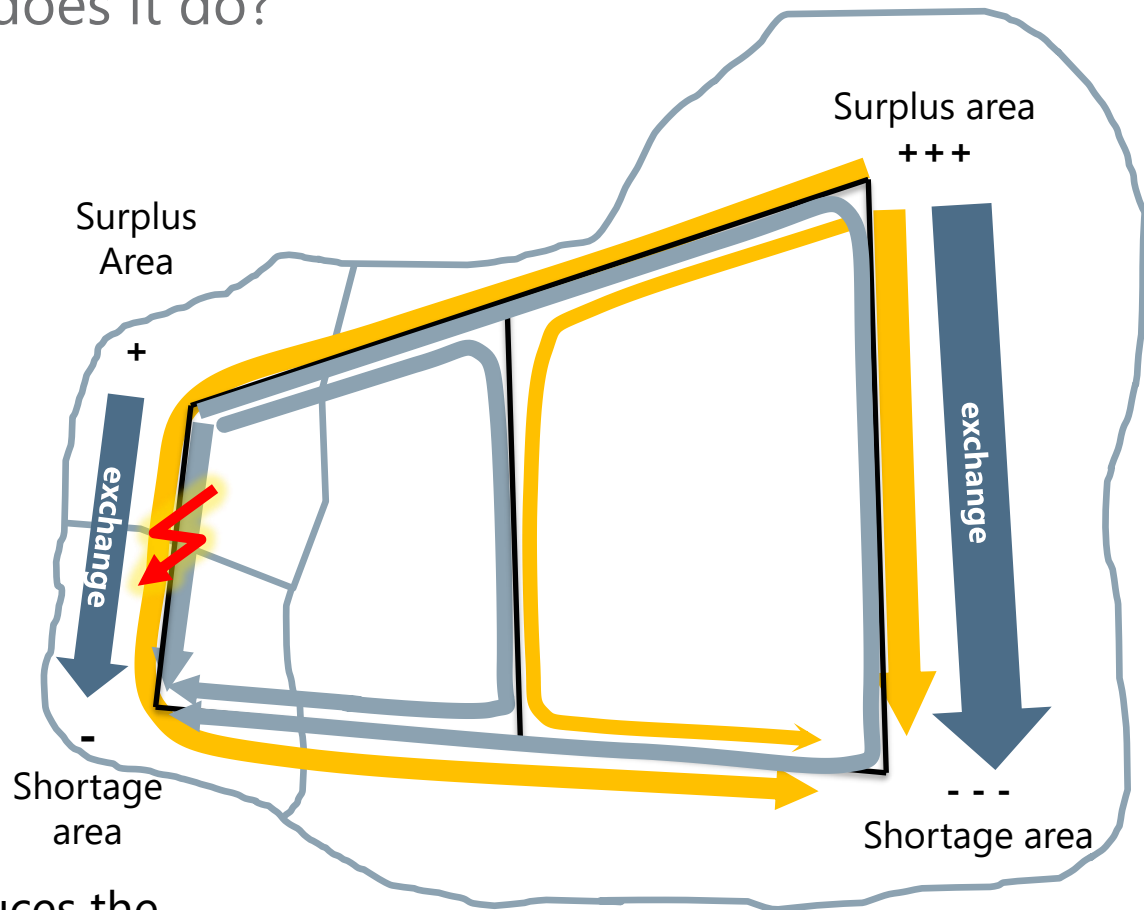
Zone delineation: what does it do?

Which exchange is causing the congestion?

- ➔ Flow caused by exchange not under allocation
- ➔ Flow caused by exchange under allocation

(Both are, but only one falls under allocation due to the current zone delineation)

The allocation mechanism reduces the exchange under allocation until the congestion is relieved



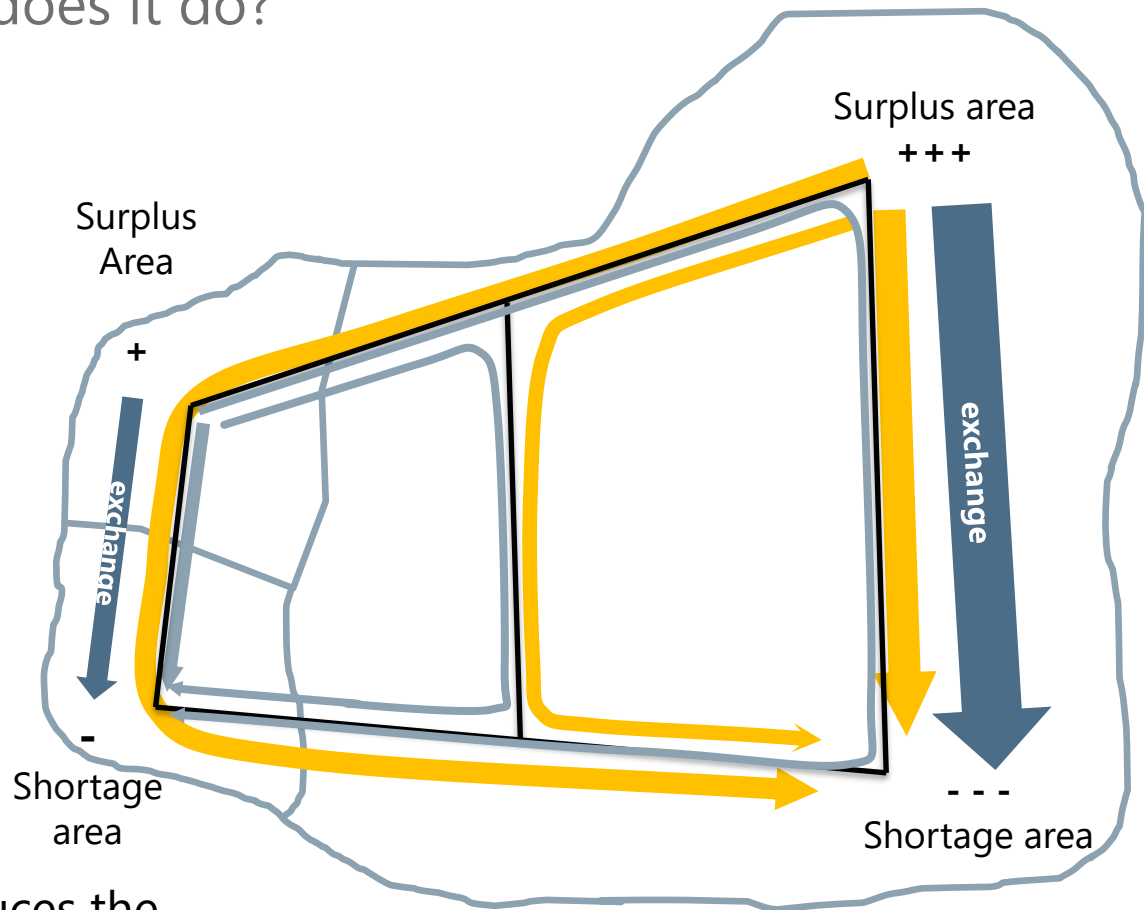
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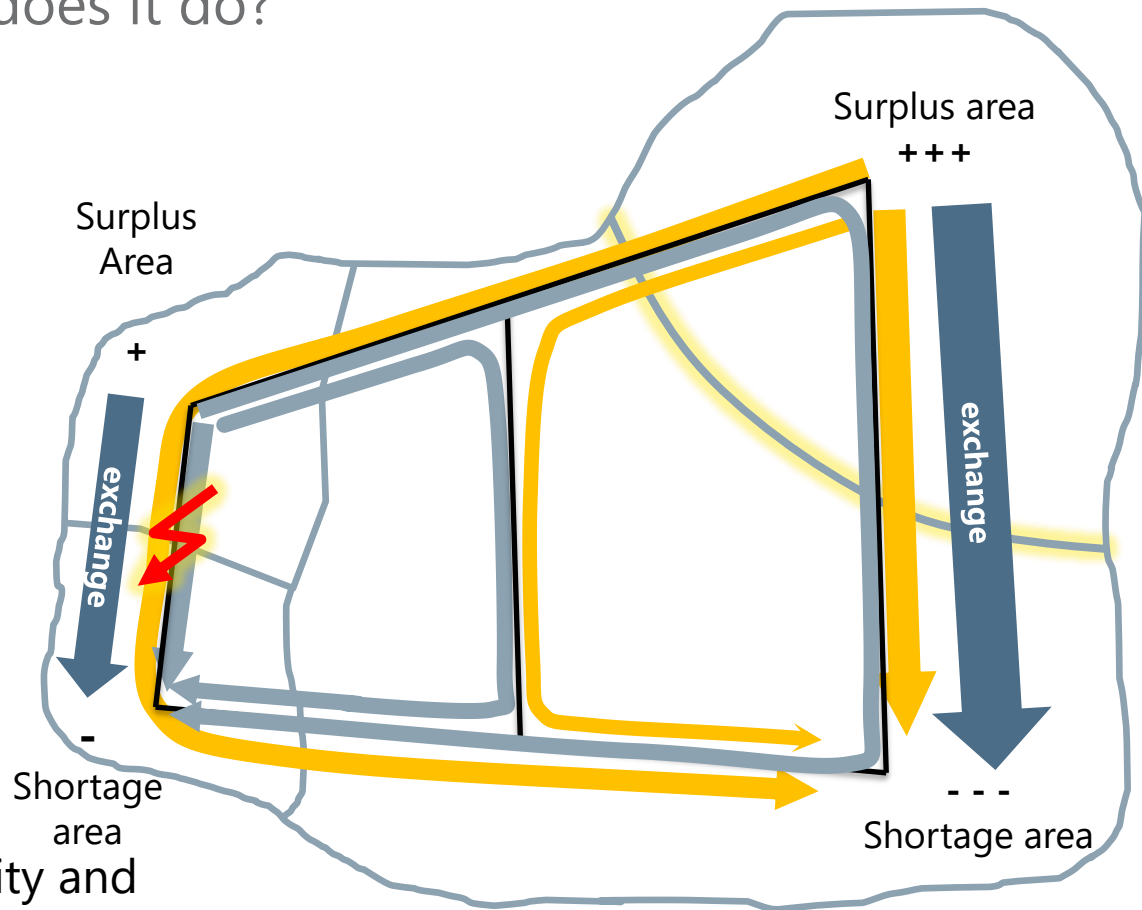


Zone delineation: what does it do?

- ➔ Flow caused by exchange not under allocation
- ➔ Flow caused by exchange under allocation

A new bidding zone brings the other exchange also under allocation

Both exchanges now compete for the scarce capacity and the price differences determine the capacity distribution between the two exchanges

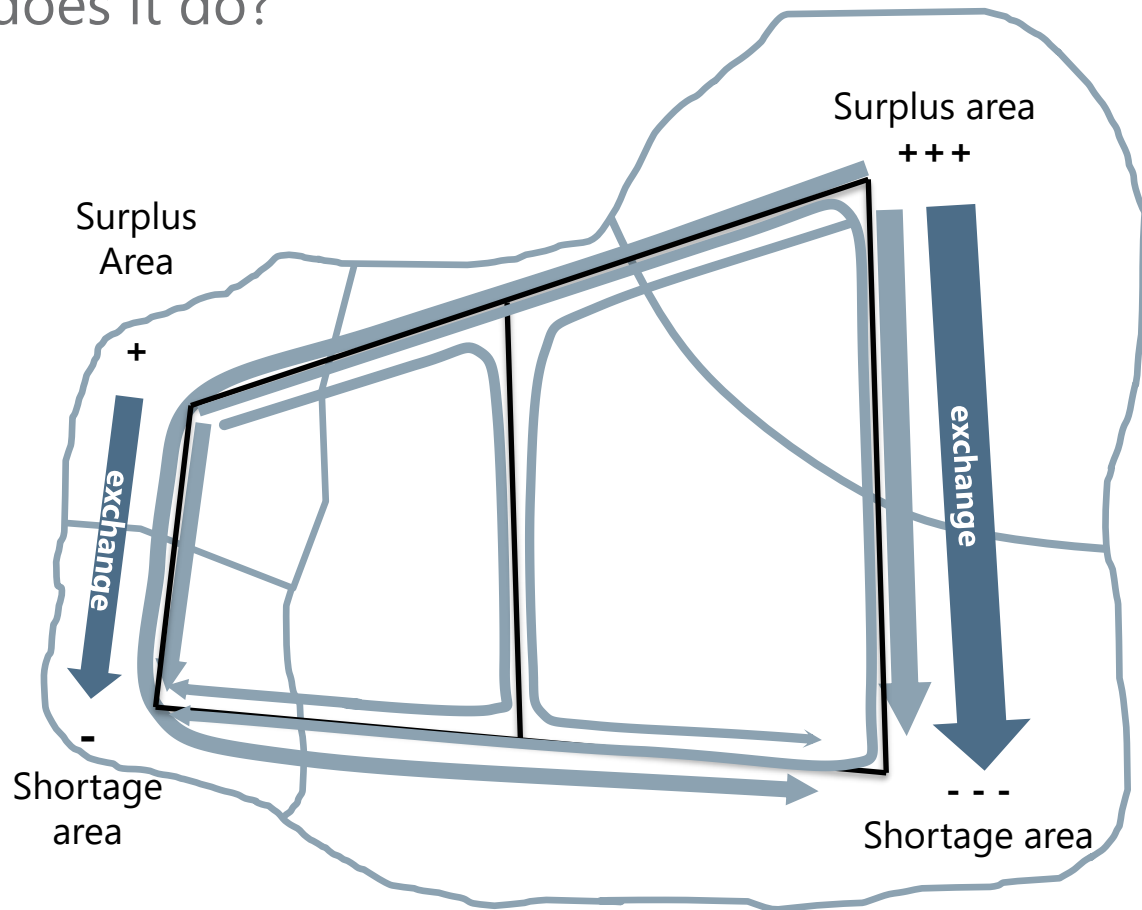


Zone delineation: what does it do?

- ➡ Flow caused by exchange not under allocation
- ➡ Flow caused by exchange under allocation

Both exchanges now fall under allocation

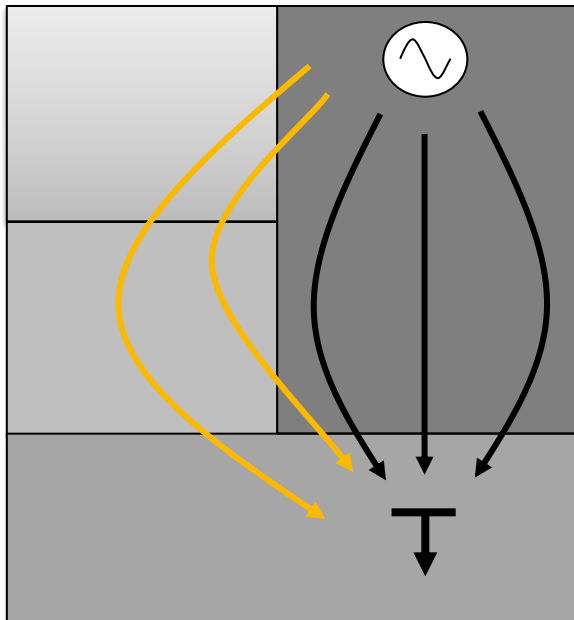
The congestion is efficiently managed through the allocation



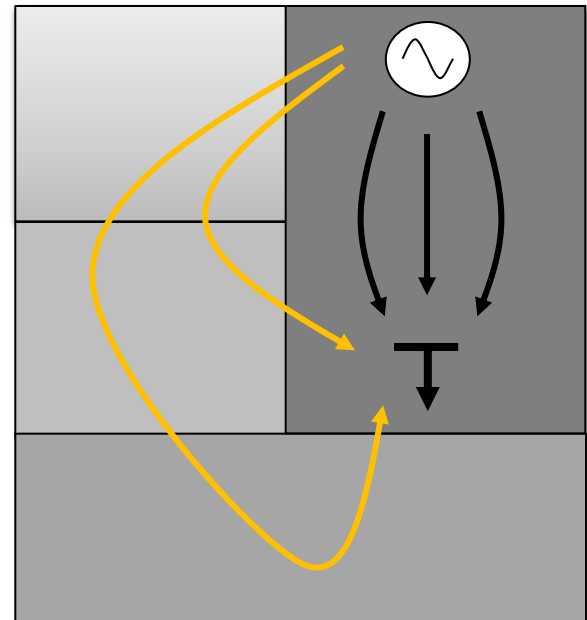
Loop flow and transit flow

- a geographical concentration of sources and/or sinks can have a strong impact on the physical flows
 - transit flows
 - loop flows

Transit flow: sink and source in different countries



Loop flow: sink and source in same country



Examples of commercial and physical flows

Wednesday December 22nd 2004 18h30

Flows:

D → NL 4557 MW

B ← NL 1540 MW

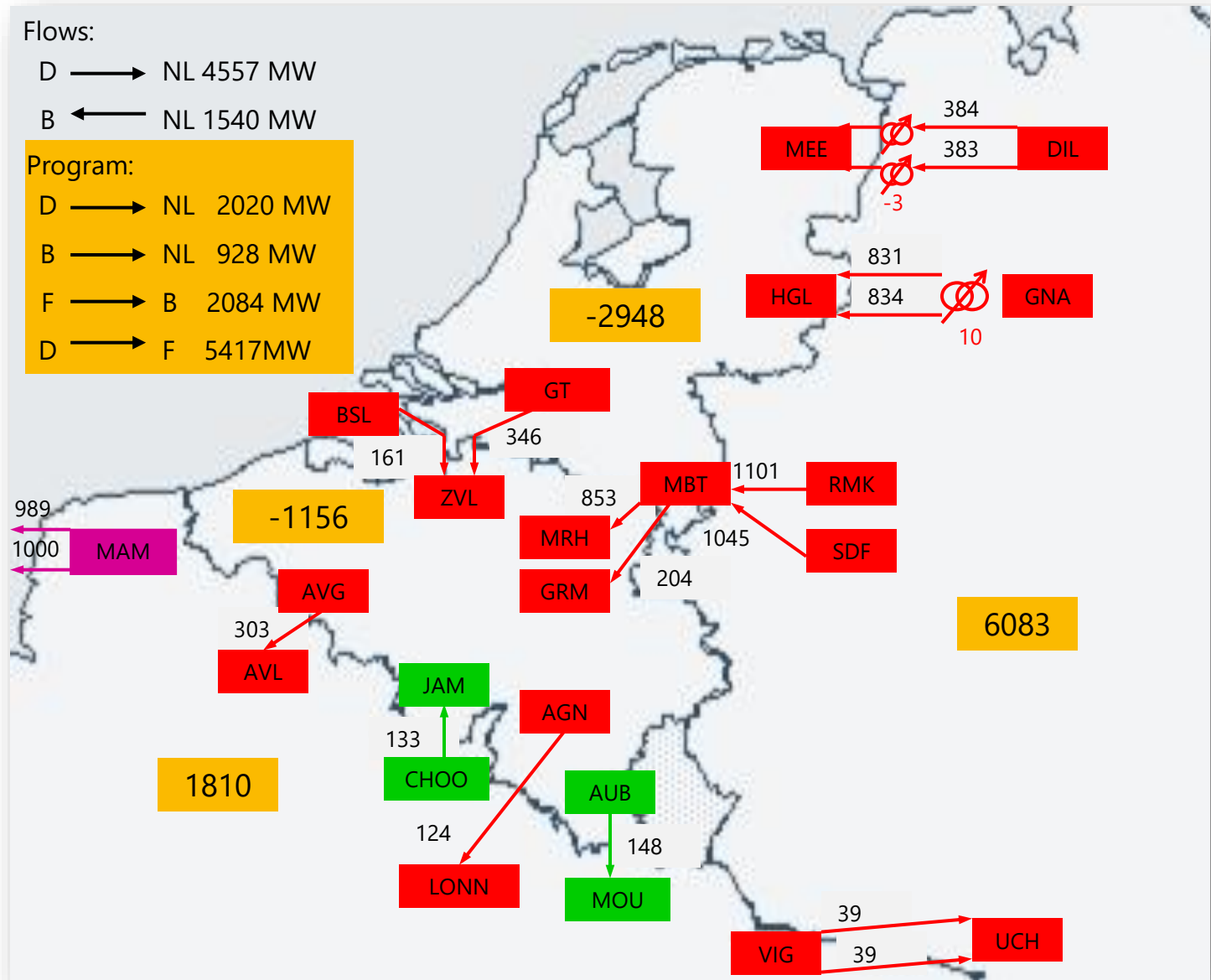
Program:

D → NL 2020 MW

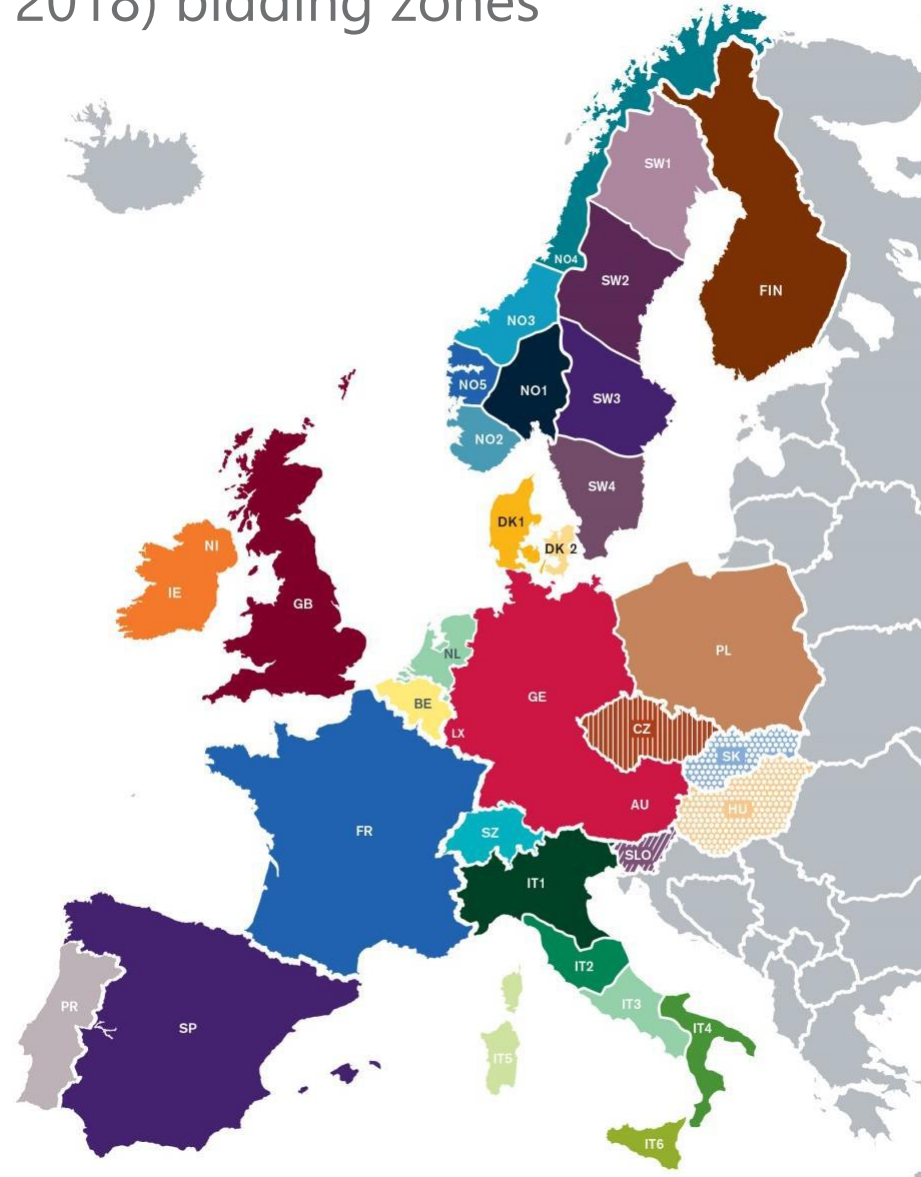
B → NL 928 MW

F → B 2084 MW

D → F 5417 MW



Old (before Oct 2018) bidding zones in Europe



Bidding zone delineation is a hot topic in Europe (1)

The Czech Republic, Hungary, Poland and Slovakia and their transmission system operators respectively have been seriously affected by recent developments in the European transmission system. Great concern has been caused by one specific serious impact on transmission systems – has taken attention.

What is concerning the public media that CEPS, SEPS, PSE and MAVIR have issued a report where they have referred to the congestion as a “necessary pre-condition for solving unplanned power flows”. Unfortunately, these power flows are referring also to the German-Austrian market zone and they may create decisions that will affect the borders of the North.

Thank you for your letter dated 3 April 2012. I must admit that I was deeply concerned by the attitude of the letter.

We read with high disconcert that you mentioned that the zone should be responsible for endangering the security of the extraordinary situation which occurred during Winter 2011, with weather and low temperatures) in the South Europe and large winter.

Additionally we are irritated by your statements,

Finally we read with anger

Bidding zone delineation is a hot topic in Europe (2)



OPINION OF THE AGENCY FOR THE COOPERATION OF ENERGY REGULATORS No 09/2015

of 23 September 2015

ON THE COMPLIANCE OF NATIONAL REGULATORY AUTHORITIES' DECISIONS APPROVING THE METHODS OF ALLOCATION OF CROSS- BORDER TRANSMISSION CAPACITY IN THE CENTRAL-EAST EUROPE REGION WITH REGULATION (EC) No 714/2009 AND THE GUIDELINES ON THE MANAGEMENT AND ALLOCATION OF AVAILABLE TRANSFER CAPACITY OF INTERCONNECTIONS BETWEEN NATIONAL SYSTEMS CONTAINED IN ANNEX I THERETO

- (94) Currently, the cross-border exchanges on the DE-AT border, while having a significant impact on structural congestions in the CEE region, do not compete for the limited capacity of these congested network elements.
- (97) In the Agency's view, the inclusion of the DE-AT border in a capacity allocation procedure coordinated within the CEE region, pursuant to points 3.1 and 3.5 of the Guidelines, would be the most appropriate market-based congestion management method enhancing competition, as it would ensure that all requests for cross-border exchanges are granted or denied based on their monetary value weighed against their relative impact on congestion.

Bidding zone delineation is a hot topic in Europe (3)



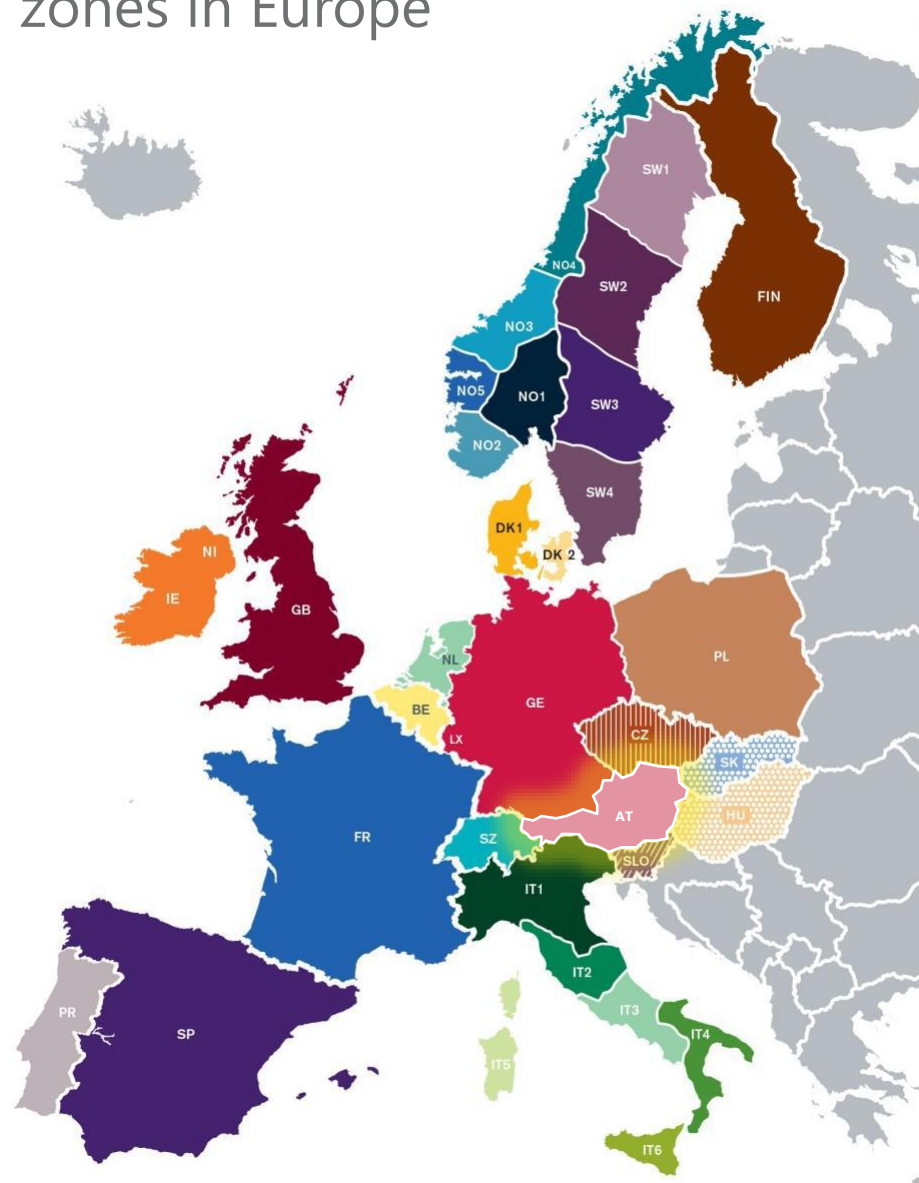
DECISION OF THE AGENCY FOR THE COOPERATION OF ENERGY REGULATORS No 06/2016

of 17 November 2016

ON THE ELECTRICITY TRANSMISSION SYSTEM OPERATORS' PROPOSAL FOR THE DETERMINATION OF CAPACITY CALCULATION REGIONS

- (62) Therefore, the CCRs Proposal may and shall include a bidding zone border between Germany/Luxembourg and Austria in defining the bidding zone borders attributed to the TSOs who are members of each CCR in accordance with Article 15(2) of the CACM Regulation.

Current bidding zones in Europe



What's next?

L 158/54

EN

Official Journal of the European Union

14.6.2019

REGULATION (EU) 2019/943 OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL

of 5 June 2019

on the internal market for electricity

(recast)

**"Clean Energy
Package"**

'Transmission system operators shall not limit the volume of interconnection capacity to be made available to market participants as a means of solving congestion inside their own bidding zone or as a means of managing flows resulting from transactions internal to bidding zones. Without prejudice to the application of the derogations under paragraphs 3 and 9 of this Article and to the application of Article 15(2), this paragraph shall be considered to be complied with where the following minimum levels of available capacity for cross-zonal trade are reached:

(a) for borders using a coordinated net transmission capacity approach, the minimum capacity shall be 70 % of the transmission capacity respecting operational security limits after deduction of contingencies, as determined in accordance with the capacity allocation and congestion management guideline adopted on the basis of Article 18(5) of Regulation (EC) No 714/2009;'

...

The total amount of 30 % can be used for the reliability margins, loop flows and internal flows on each critical network element.



Legislation to incentivize splitting of "too large" bidding zones

What's next?



ACER Decision on the methodology and assumptions that are to be used in the bidding zone review process and for the alternative bidding zone configurations to be considered: Annex I

Methodology and assumptions that are to be used in the bidding zone review process

in accordance with Article 14(5) of the Regulation (EU) 2019/943 of the European Parliament and of the Council of 5 June 2019 on the internal market for electricity

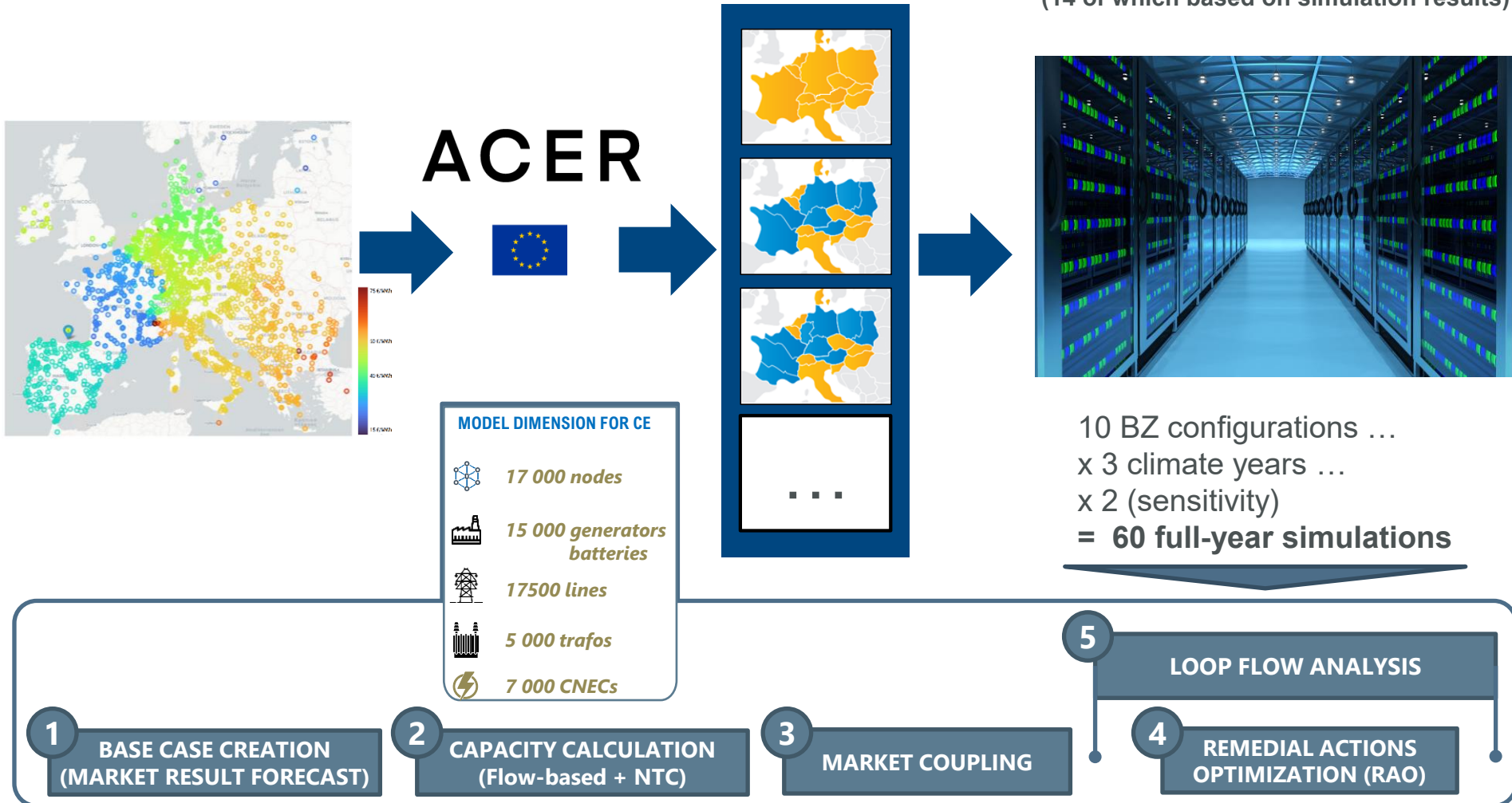


Main study phase...the process

Locational Marginal
Prices (LMPs)

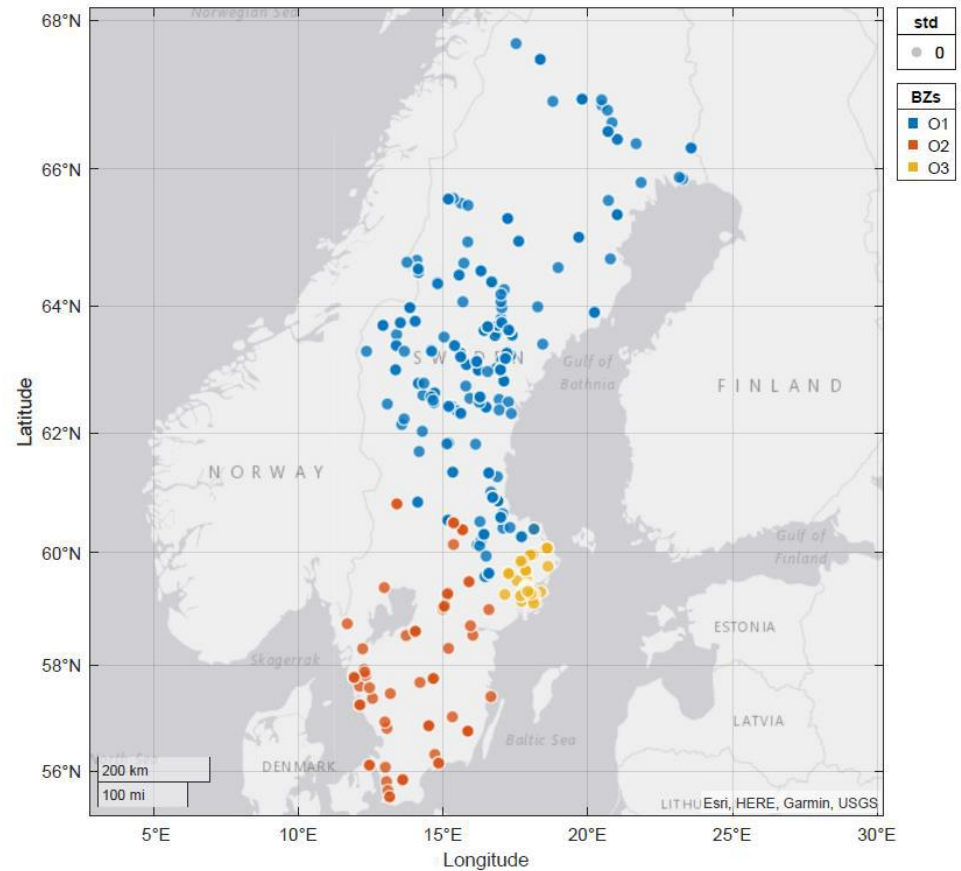
Up to 10
configurations

Evaluation based on
22 indicators
(14 of which based on simulation results)



Bidding zone configurations to be assessed

Identifier	BZRR	Number of BZs per MS
1	CE	DE2
2		
12	CE	DE3
13	CE	DE4
14	CE	DE5
5	CE	FR3
6	CE	IT2
7	CE	NL2
8	Nordic	SE3
9	Nordic	SE3
10	Nordic	SE4
11	Nordic	SE4



Key findings of the Bidding Zone Review for the year 2025

The results of the economic efficiency criterion of alternative BZs are the following:

- For the **Nordic** Region: None of the studied alternative BZ configurations provide a higher economic efficiency. The result shows a negative change in economic efficiency for the configurations ranging from €2 million to €35 million compared to the status quo.
 - The Nordic TSOs propose to **maintain the current BZ configuration in Sweden**
- For the **Central Europe Region**:
 - The simulation results show a higher economic efficiency for all German–Luxembourgish split configurations (ranging from €251 million to €339 million for the 2025 target year), where the split of Germany–Luxembourg into five BZs performs the highest among the analysed alternative configurations in terms of economic efficiency.
 - The Dutch split configuration also shows a slight positive effect on economic efficiency (€9 million for the 2025 target year).
 - The French and Italian alternative configurations show a negative effect in economic efficiency.
 - For the TSOs of the Central Europe region, the simulation results indicate that the **configuration with the highest positive monetised benefit in relation to criterion four (economic efficiency) compared to the status quo would be the split of Germany–Luxembourg into five BZs**. The Central Europe TSOs proposal stresses that this result stems from the BZR methodology defined by ACER and does not take important additional aspects into account and therefore should not be seen in isolation, but rather in combination with certain considerations, which should be **thoroughly assessed prior to the eventual decision of the relevant member state(s) affected by a split on the future BZ configuration**, as they could have a considerable impact on the interpretation and the outcomes of the BZ Study performed by the TSOs.

“Germany remains committed to its single bidding zone”

“The Federal Republic of Germany remains committed to the existing single German-Luxembourg bidding zone. This decision is also in line with the current coalition agreement between the CDU/CSU and SPD parties, which reaffirms the commitment to the single bidding zone. By maintaining the single bidding zone, a uniform exchange electricity price will continue to apply in Germany. In terms of electricity prices, this means that all German consumers will also benefit in future from the high liquidity of the German electricity market, i.e. the possibility of buying and selling electricity trading products for different delivery horizons in large quantities. In addition, by maintaining the single bidding zone, German consumers will continue to benefit from the best generation conditions within the large German-Luxembourg electricity bidding zone.

From the Federal Government’s point of view, the results of the bidding zone review represent only a snapshot, as it only takes the year 2025 into consideration. **The prosperity gains simulated in this case for the year 2025 are far lower than the estimated adjustment costs that would result from amending the bidding zone configuration, which means that the new bidding zones would have to remain in place for many years in order to be profitable.** In addition, **the German transmission grid will be significantly upgraded and strengthened in the years to come.** The commissioning of the first intra-German high-voltage direct current (HVDC) transmission lines, for example, will increase the transport capacity of the German transmission grid considerably and reduce current bottlenecks. **For this and other reasons, it is highly doubtful therefore whether it will be possible to achieve the simulated prosperity gains at all.**

For reasons beyond its control, Germany also rejects an amendment of the electricity bidding zone configuration, as this would increase investment uncertainty in the energy industry significantly and would lead to considerable regional cost differences for end consumers. It would also call the economic viability of generation plants in some regions into question and would raise complex industrial policy issues at a time when European industry is already facing fundamental challenges.”

Phase shifting transformers in Europe



This map is without prejudice to the status of or sovereignty over any territory, to the delimitation of international borders and boundaries and to the name of any territory, city or area.

Note: The phase-shifters in light pink colour are under preparation or analysis.

* Phase-shifters are planned in these countries.

Source: IEA adapted on the basis of data by ACER/CEER (2013), *ACER/CEER Annual Report on the Results of Monitoring the Internal Electricity and Natural Gas Markets in 2012*, November, Ljubljana/Brussels.

Any
Questions

In case of questions, feel free to contact me!



E-Bridge Consulting B.V.
Utrechtseweg 159a
6862 AH Oosterbeek
The Netherlands

phone +31 (0) 26 7009791
Mobile +31 (0) 6 42263895
mail pschavemaker@e-bridge.com
Web www.e-bridge.com



Dr. Pieter Schavemaker
Managing Director

E-Bridge Consulting B.V.

Utrechtseweg 159a

6862 AH Oosterbeek, the Netherlands

Phone +31 (0)26 700 9797

E-mail info@e-bridge.nl

For more information about our
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